Problem Statement:

Recruiting and retaining drivers is seen by industry watchers as a tough battle for Ola. Churn among drivers is high and it’s very easy for drivers to stop working for the service on the fly or jump to Uber depending on the rates.

As the companies get bigger, the high churn could become a bigger problem. To find new drivers, Ola is casting a wide net, including people who don’t have cars for jobs. But this acquisition is costly. Losing drivers frequently impacts the morale of the organization and acquiring new drivers is more expensive than retaining existing ones.

Column Profiling:

1. MMMM-YY : Reporting Date (Monthly)
2. Driver_ID : Unique id for drivers
3. Age : Age of the driver
4. Gender : Gender of the driver – Male : 0, Female: 1
5. City : City Code of the driver
6. Education_Level : Education level – 0 for 10+ ,1 for 12+ ,2 for graduate
7. Income : Monthly average Income of the driver
8. Date Of Joining : Joining date for the driver
9. LastWorkingDate : Last date of working for the driver
10. Joining Designation : Designation of the driver at the time of joining
11. Grade : Grade of the driver at the time of reporting
12. Total Business Value : The total business value acquired by the driver in a month (negative business indicates cancellation/refund or car EMI adjustments)
13. Quarterly Rating : Quarterly rating of the driver: 1,2,3,4,5 (higher is better)

Import the necessary libraries & Basic Analysis

In [ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
df = pd.read_csv("https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/0
```

In [ ]:
```
df.head()
```

```
/usr/local/lib/python3.12/dist-packages/google/colab/_dataframe_summarizer.py:8
8: UserWarning: Could not infer format, so each element will be parsed individu
ally, falling back to `dateutil`. To ensure parsing is consistent and as-expect
ed, please specify a format.
  cast_date_col = pd.to_datetime(column, errors="coerce")
```

Out[ ]:

	Unnamed: 0	MMM-YY	Driver_ID	Age	Gender	City	Education_Level	Income	Da
0	0	01/01/19	1	28.0	0.0	C23	2	57387	
1	1	02/01/19	1	28.0	0.0	C23	2	57387	
2	2	03/01/19	1	28.0	0.0	C23	2	57387	
3	3	11/01/20	2	31.0	0.0	C7	2	67016	
4	4	12/01/20	2	31.0	0.0	C7	2	67016	

Exploratory Data Analysis

In [ ]:
```
df.shape
```

Out[ ]: (19104, 14)

- The DataSet Contains 19104 rows and 14 columns

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19104 entries, 0 to 19103
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Unnamed: 0            19104 non-null  int64
 1   MMM-YY                19104 non-null  object
 2   Driver_ID             19104 non-null  int64
 3   Age                   19043 non-null  float64
 4   Gender                19052 non-null  float64
 5   City                  19104 non-null  object
 6   Education_Level       19104 non-null  int64
 7   Income                19104 non-null  int64
 8   Dateofjoining         19104 non-null  object
 9   LastWorkingDate       1616 non-null   object
 10  Joining Designation   19104 non-null  int64
 11  Grade                 19104 non-null  int64
 12  Total Business Value  19104 non-null  int64
 13  Quarterly Rating      19104 non-null  int64
dtypes: float64(2), int64(8), object(4)
memory usage: 2.0+ MB
```

In [ ]:
```
# check unique values in categorical variables
for col in df:
    print("Column:  {:18}   Unique values: {}".format(col, df[col].nunique()))
```

```
Column:  Unnamed: 0           Unique values: 19104
Column:  MMM-YY               Unique values: 24
Column:  Driver_ID            Unique values: 2381
Column:  Age                  Unique values: 36
Column:  Gender               Unique values: 2
Column:  City                 Unique values: 29
Column:  Education_Level      Unique values: 3
Column:  Income               Unique values: 2383
Column:  Dateofjoining        Unique values: 869
Column:  LastWorkingDate      Unique values: 493
Column:  Joining Designation   Unique values: 5
Column:  Grade                Unique values: 5
Column:  Total Business Value   Unique values: 10181
Column:  Quarterly Rating     Unique values: 4
```

In [ ]:
```
df.describe()
```

Out[ ]:

	Unnamed: 0	Driver_ID	Age	Gender	Education_Level
count	19104.000000	19104.000000	19043.000000	19052.000000	19104.000000
mean	9551.500000	1415.591133	34.668435	0.418749	1.021671
std	5514.994107	810.705321	6.257912	0.493367	0.800167
min	0.000000	1.000000	21.000000	0.000000	0.000000
25%	4775.750000	710.000000	30.000000	0.000000	0.000000
50%	9551.500000	1417.000000	34.000000	0.000000	1.000000
75%	14327.250000	2137.000000	39.000000	1.000000	2.000000
max	19103.000000	2788.000000	58.000000	1.000000	2.000000

In [ ]:
```
df.columns
```

Out[ ]:
```
Index(['Unnamed: 0', 'MMM-YY', 'Driver_ID', 'Age', 'Gender', 'City',
       'Education_Level', 'Income', 'Dateofjoining', 'LastWorkingDate',
       'Joining Designation', 'Grade', 'Total Business Value',
       'Quarterly Rating'],
      dtype='object')
```

Separating Numerical variables and Categorical variables.

In [ ]:
```
num_cols = df.select_dtypes(include=['int64','float64']).columns.tolist()
cat_cols = df.select_dtypes(include=['object']).columns.tolist()
for col in cat_cols:
    df[col] = df[col].astype('category')
```

Making the target variable based on the leaving date of the driver

In [ ]:
```
df['LastWorkingDate'].isna().sum()
```

Out[ ]:
```
np.int64(17488)
```

In [ ]:
```
df['Target'] = df['LastWorkingDate'].apply(lambda x: 0 if pd.notna(x) else
1)
df.drop('LastWorkingDate', axis = 1, inplace = True)
df['Target'].value_counts()
```

Out[ ]:

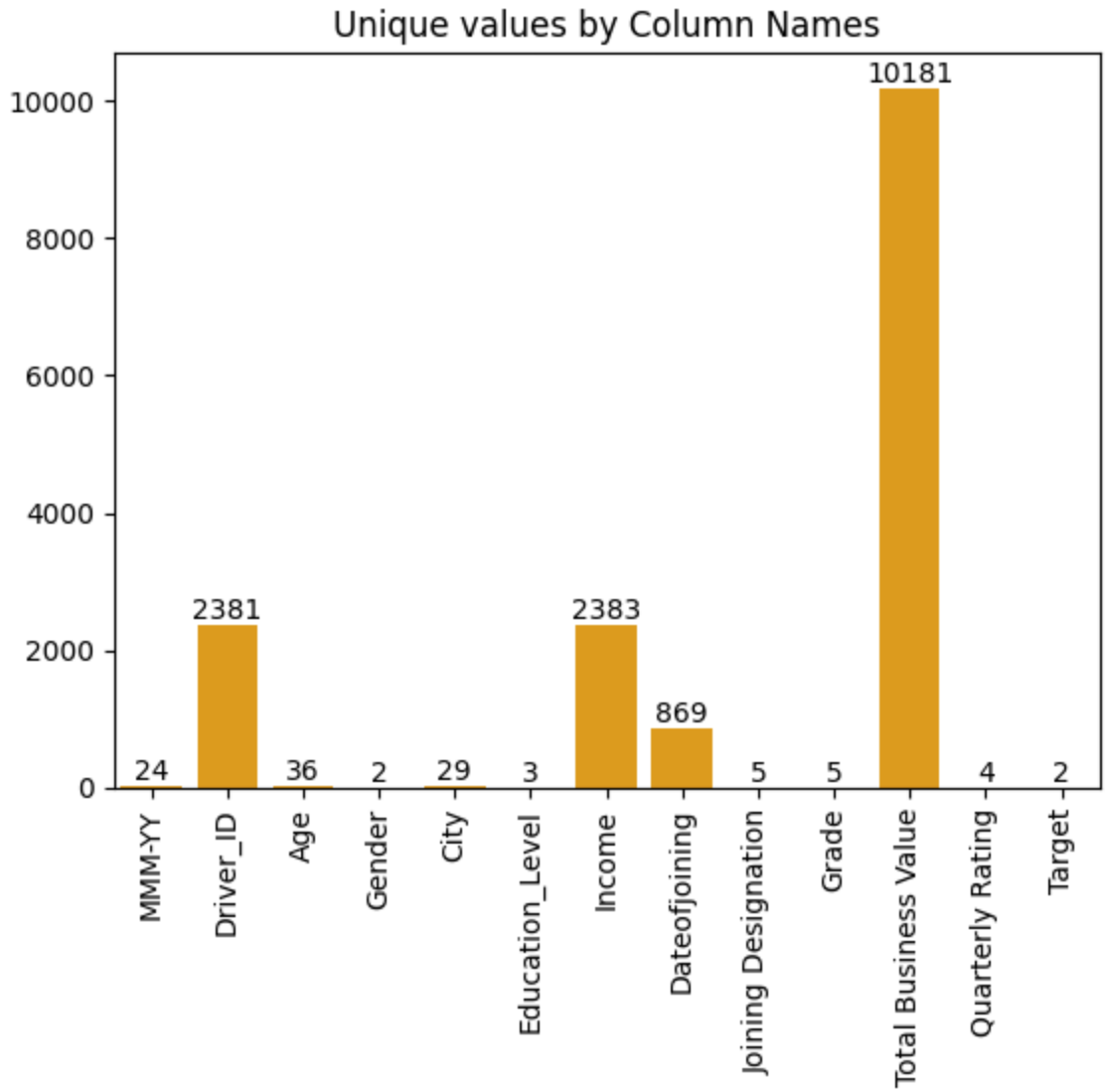
Target	count
1	17488
0	1616

dtype: int64

As the Unnamed 0 Column is not providing any useful information so dropping it.

In [ ]:

```
# Removing the unwanted field 'Unnamed : 0
df.drop('Unnamed: 0',axis = 1,inplace=True)

# Checking the unique count in each column and total count
col_name = [cols for cols in df.columns]
unique_count = [df[cols].nunique() for cols in df.columns]

gr = sns.barplot(y = unique_count,x = col_name,color='orange')
gr.bar_label(gr.containers[0])
plt.title('Unique values by Column Names')
plt.xticks(rotation = 90)
plt.show()
```



Missing value detection

In [ ]:
```
df.isnull().sum()
```

Out[ ]:

	0
MMM-YY	0
Driver_ID	0
Age	61
Gender	52
City	0
Education_Level	0
Income	0
Dateofjoining	0
Joining Designation	0
Grade	0
Total Business Value	0
Quarterly Rating	0
Target	0

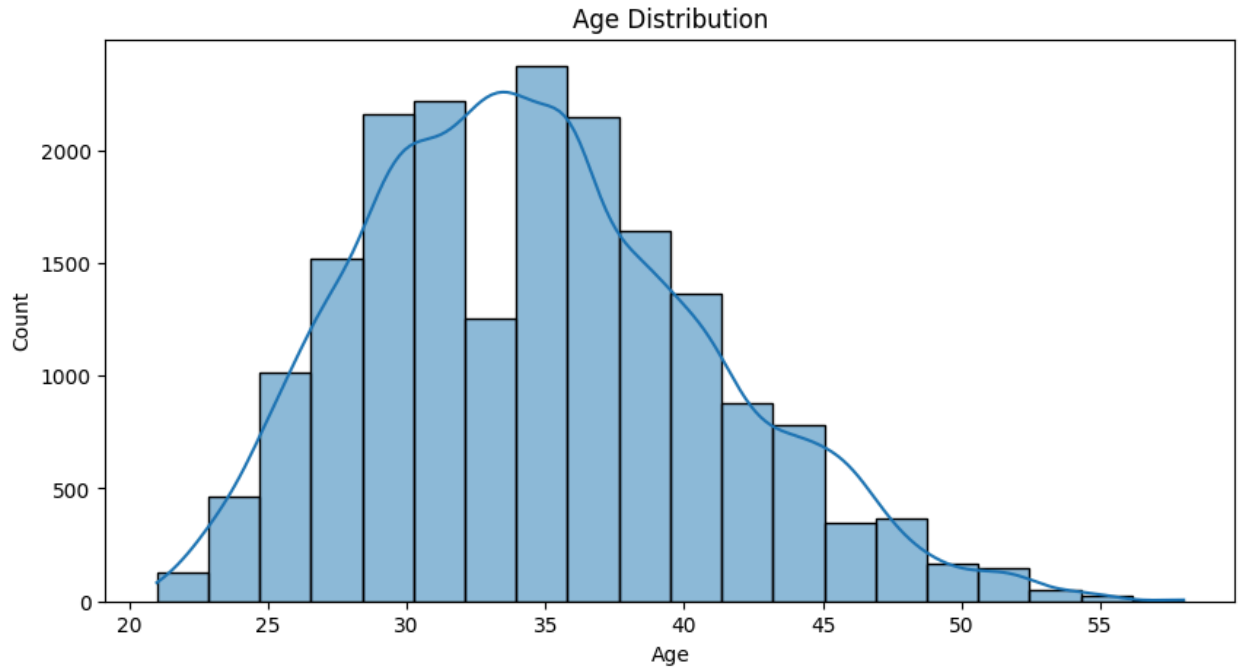
dtype: int64

- There are 61 missing values in Age column and 52 missing values in Gender column

Univariate Data Analysis

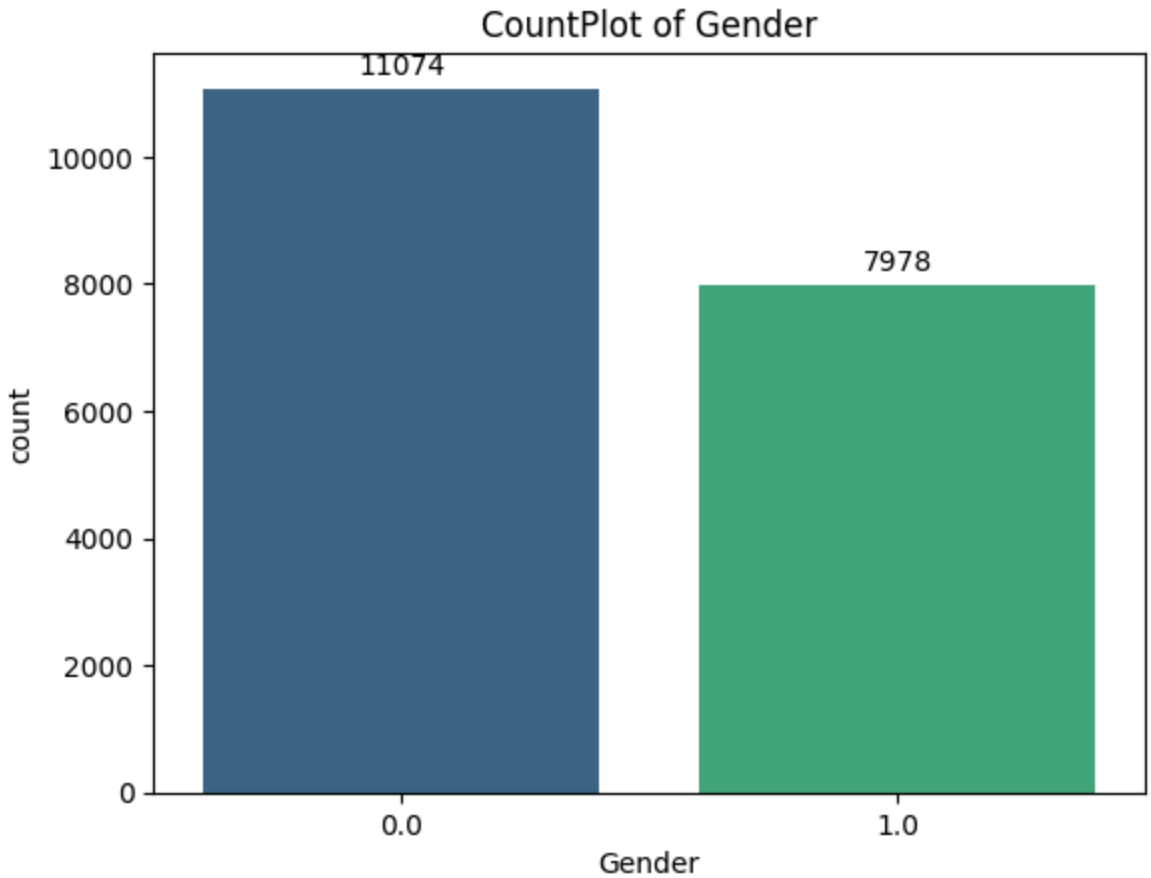
In [ ]:

```
plt.figure(figsize=(10,5))
sns.histplot(data=df, x='Age', bins=20, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



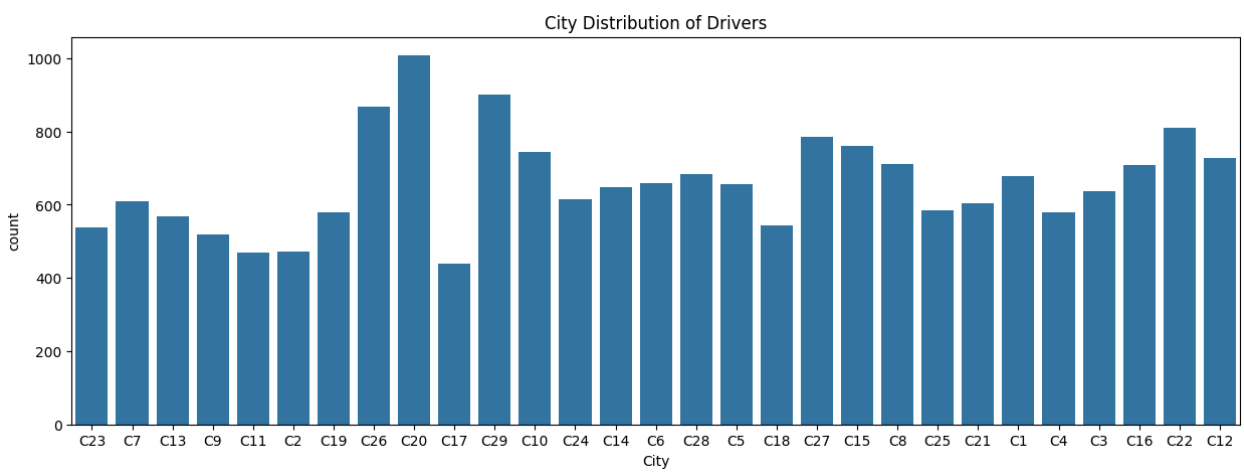
- The dataset is dominated by young to mid-age employees (25–40) with a peak around early 30s, and very few older employees.

In [ ]:
```
ax = sns.countplot(data = df,x = df['Gender'],hue = 'Gender',palette='viridis'
for container in ax.containers:
  ax.bar_label(container,padding = 3)
plt.title("CountPlot of Gender")
plt.show()
```



- There are 11074 males and 7978 females

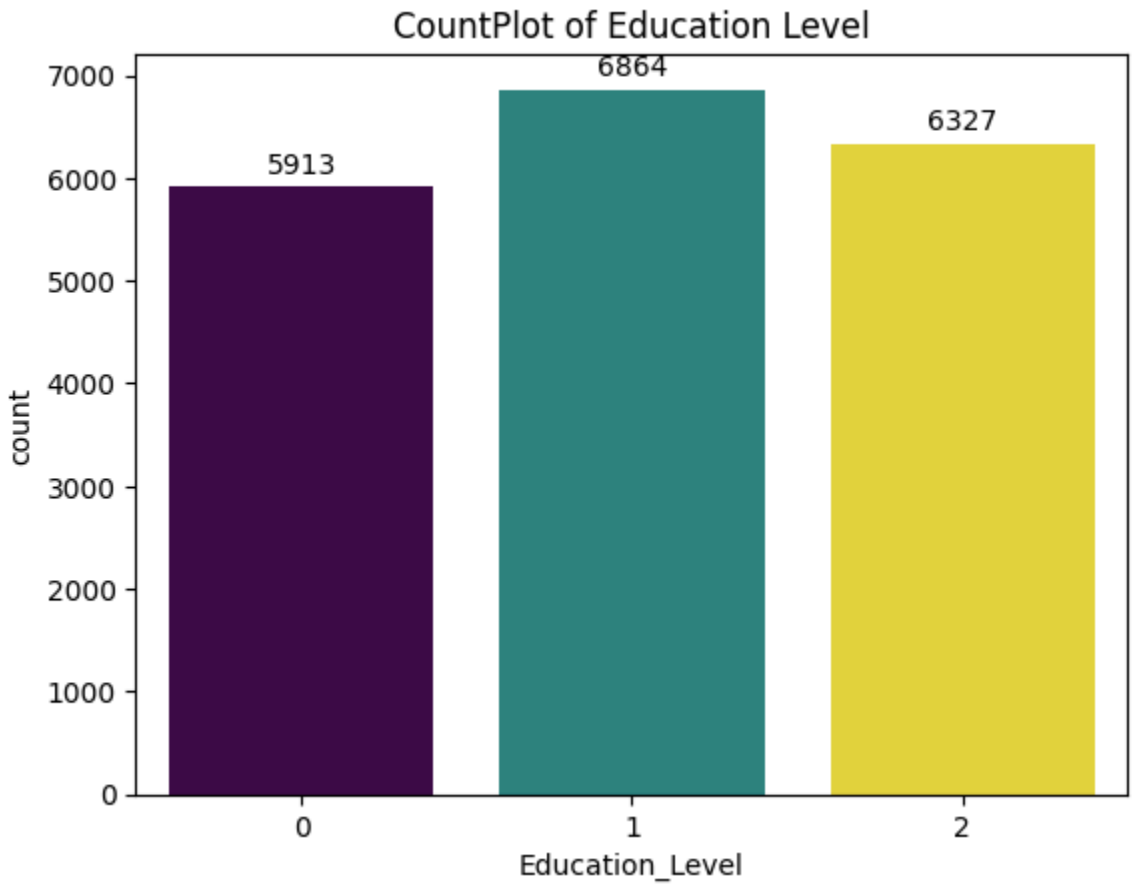
In [ ]:
```
plt.figure(figsize=(15, 5))
sns.countplot(data=df, x="City");
plt.title("City Distribution of Drivers");
```



- We can see that C20 has the highest number of drivers

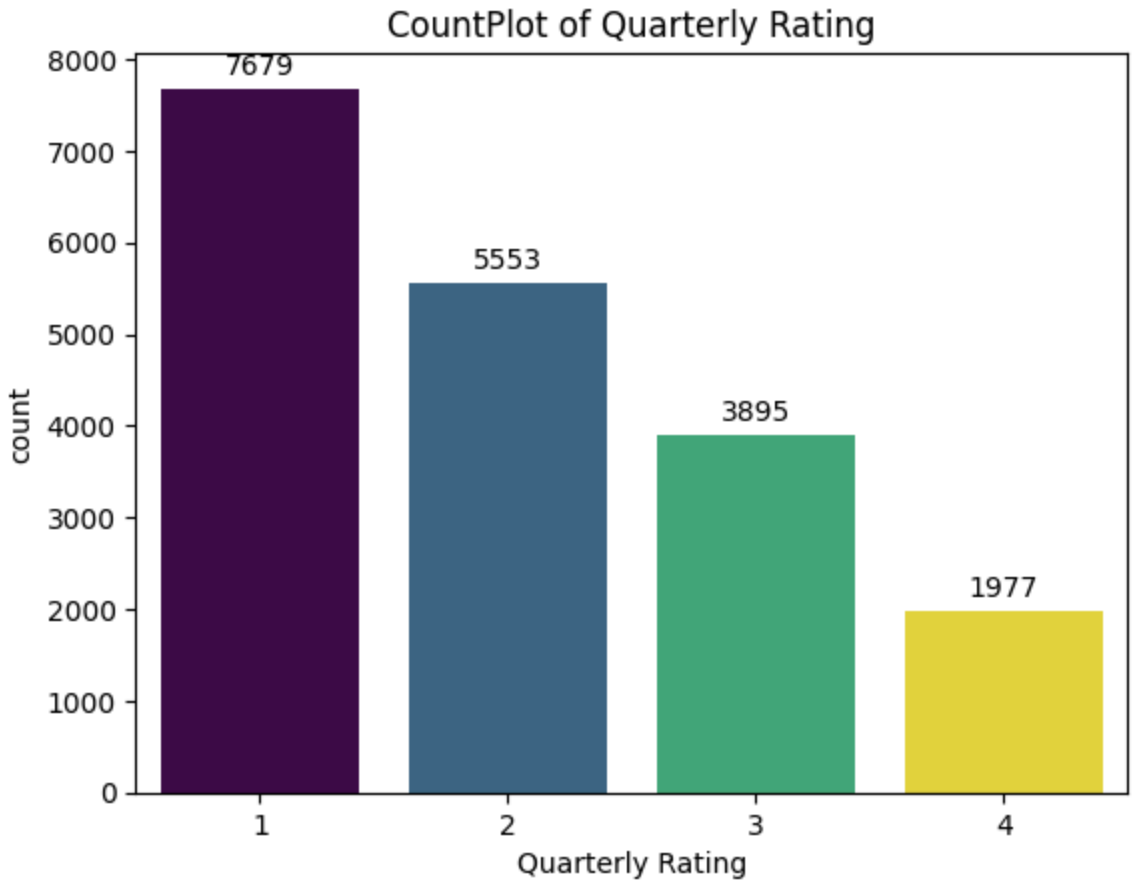
In [ ]:
```
ax = sns.countplot(data = df,x = df['Education_Level'],hue = 'Education_Level'
```

```
for container in ax.containers:
  ax.bar_label(container,padding = 3)
plt.title("CountPlot of Education Level")
plt.show()
```

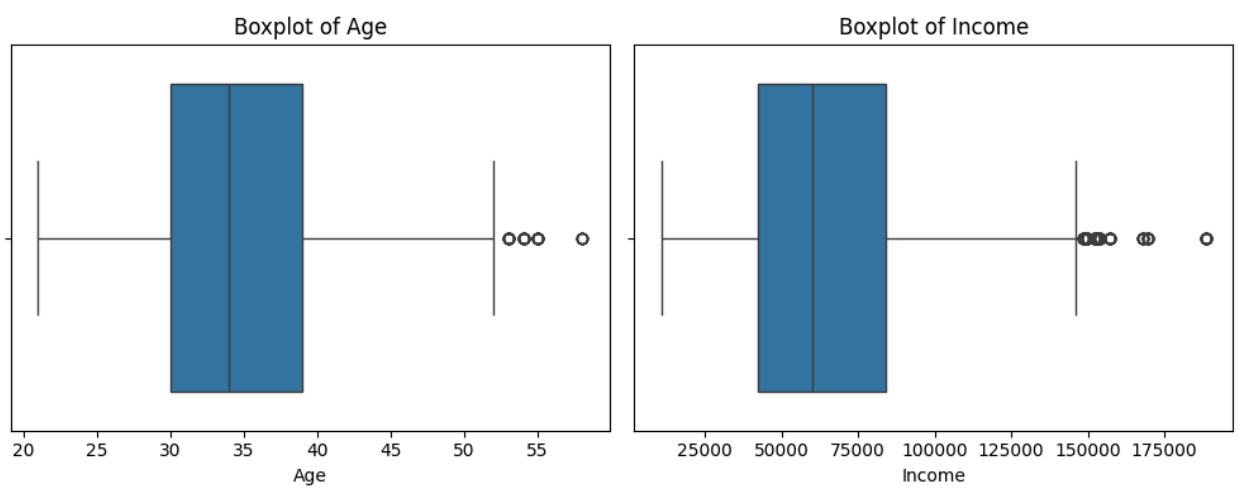


- Majority of the group are (12+)
- Graduates is the second largest group

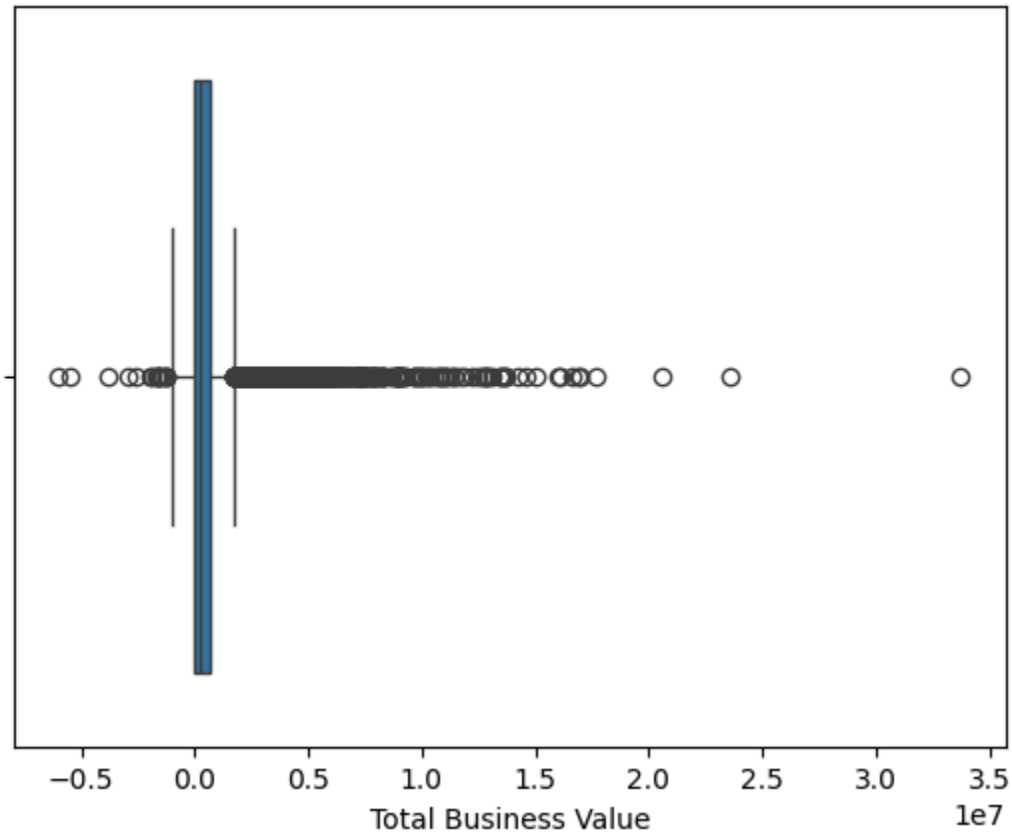
In [ ]:
```
ax = sns.countplot(data = df,x = df['Quarterly Rating'],hue = 'Quarterly Ratir
for container in ax.containers:
  ax.bar_label(container,padding = 3)
plt.title("CountPlot of Quarterly Rating")
plt.show()
```



In [ ]:
```
fig, axs = plt.subplots(1, 2, figsize=(10, 4))

sns.boxplot(x = df['Age'], ax=axs[0])
axs[0].set_title("Boxplot of Age")

sns.boxplot(x = df['Income'],ax=axs[1])
axs[1].set_title("Boxplot of Income")

plt.tight_layout()
plt.show()
```



In [ ]:
```
sns.boxplot(x = df['Total Business Value'])
plt.show()
```



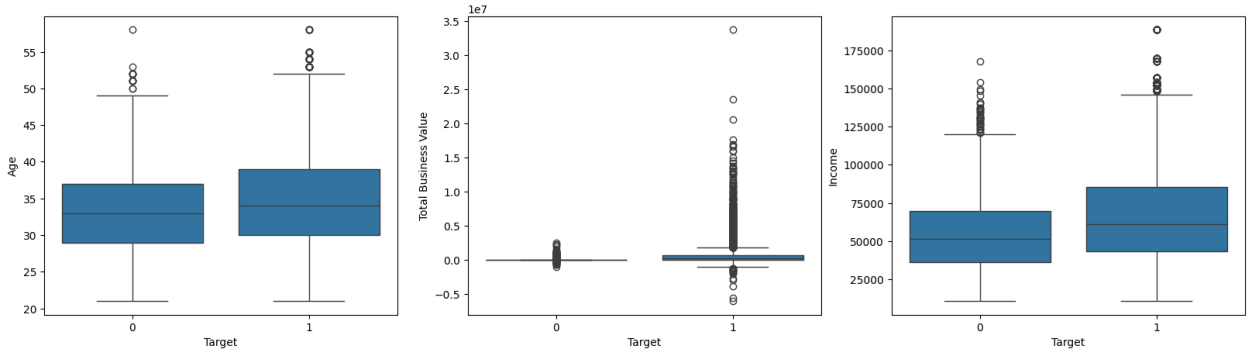
Key takeaways :

1. Median age is around mid-30s, with most employees between 30–40 Few older outliers (50+), indicating limited senior-age workforce
2. Income is right-skewed → most employees earn in lower-mid range Presence of high-income outliers, likely senior/top performers
3. Total Business Value Extremely right-skewed with heavy outliers Most values are clustered near low range, but few employees generate very high business value

Bi-variate Analysis

categorical-continuous variables

In [ ]:
```
fig, ax = plt.subplots(1, 3, figsize=(20, 5))

sns.boxplot(data=df, x="Target", y="Age", ax=ax[0]);
sns.boxplot(data=df, x="Target", y="Total Business Value", ax=ax[1]);
sns.boxplot(data=df, x="Target", y="Income", ax=ax[2]);
```

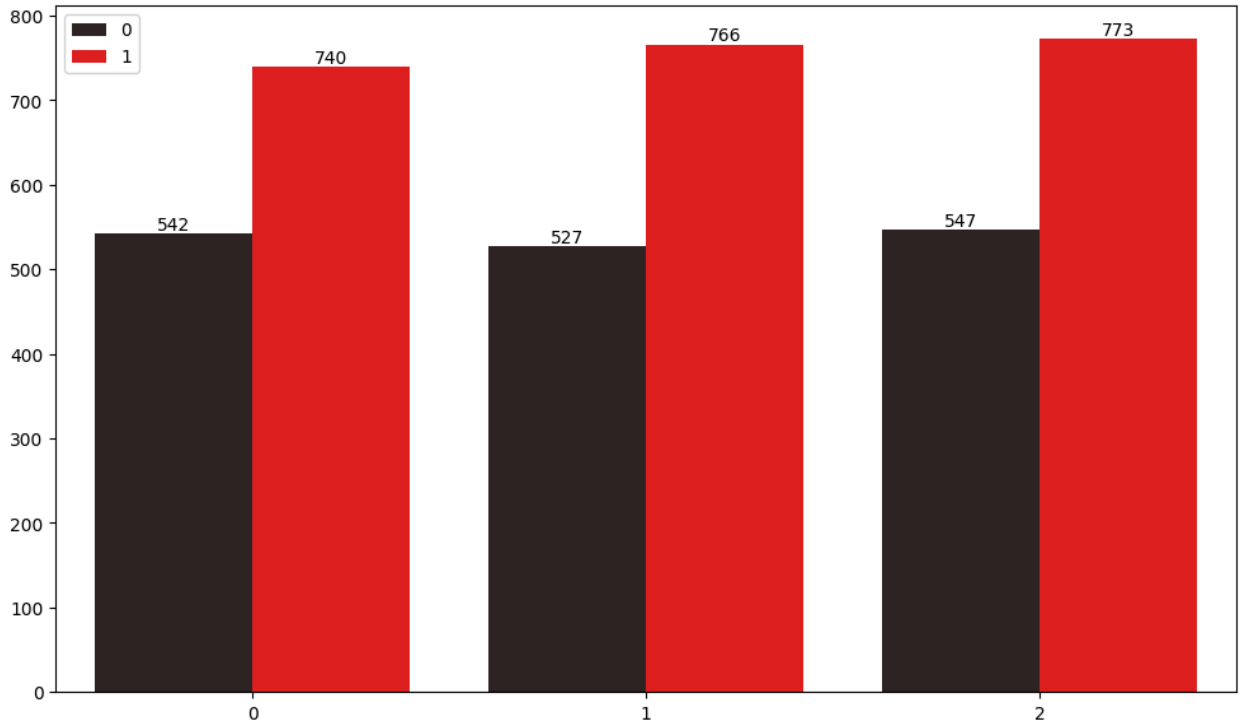


Observations

- Above plots show how different features affect the target variable
- We can see the Income, Age business value and total business value is less for churned drivers.

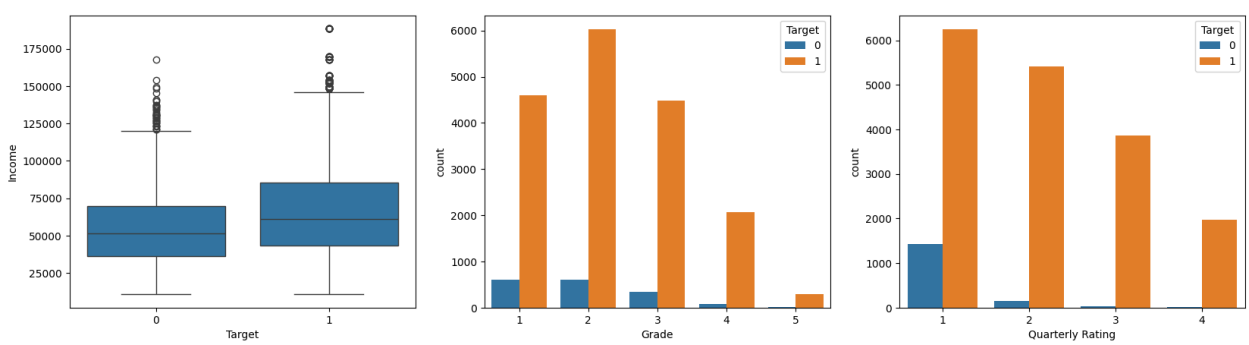
In [ ]:
```
#Here Education level – 0 for 10+ ,1 for 12+ ,2 for graduate
data = df.groupby(['Education_Level','Target'])['Driver_ID'].agg('nunique')
# plotting the graph
education_level = [tuple[0] for tuple in data.index]
target = [tuple[1] for tuple in data.index]
value = data.values
#plotting the graph
plt.figure(figsize = (12,7))
gr = sns.barplot(x = education_level, y = value, hue = target, color =
'red')
gr.bar_label(gr.containers[0])
gr.bar_label(gr.containers[1])
plt.show()
```

```
/tmp/ipykernel_784/3306298549.py:9: FutureWarning:

Setting a gradient palette using color= is deprecated and will be removed in v
0.14.0. Set `palette='dark:red'` for the same effect.

  gr = sns.barplot(x = education_level, y = value, hue = target, color =
```



- The proportion between chruned and not chruned are almost same for all the Education Levels

In [ ]:
```
fig, ax = plt.subplots(1, 3, figsize=(20, 5))
sns.boxplot(data=df, x="Target", y="Income", ax=ax[0]);
sns.countplot(data=df, x="Grade", hue="Target", ax=ax[1]);
sns.countplot(data=df, x="Quarterly Rating", hue="Target", ax=ax[2]);
```



Observations

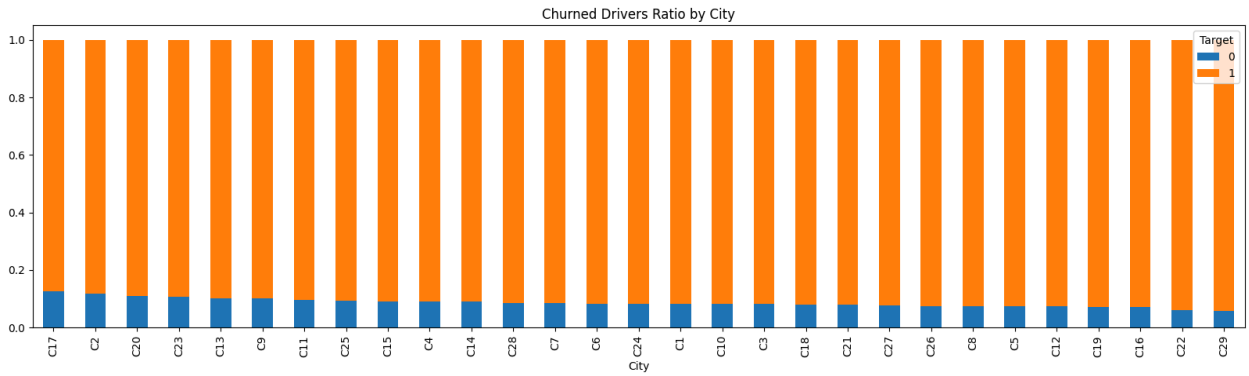
- We can see that Income is less for churned drivers as compared to current drivers.
- Low Income may be responsible for driver churn.
- There are very few Grade 4 and 5 employees as compared to Grade 1, 2 and 3.
- Majority of the drivers churning belong to Grade 1 and 2.

- Majority of the drivers who Churned had 1 rating.

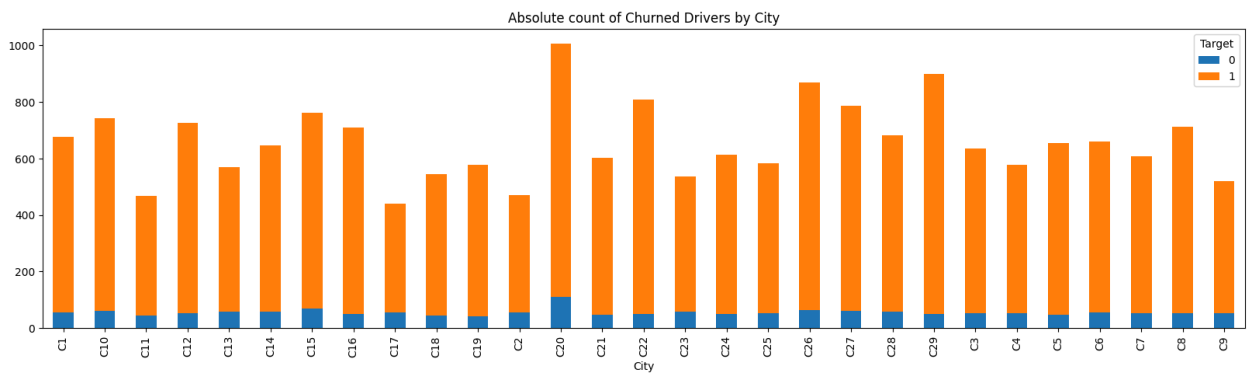
categorical-categorical Variables

Ratio of Churned drivers by City

In [ ]:
```
pd.crosstab(df["City"], df["Target"],  normalize="index").sort_values(1).plot(
plt.title("Churned Drivers Ratio by City");
```



In [ ]:
```
pd.crosstab(df["City"], df["Target"]).plot(kind="bar", stacked=True, figsize=(
plt.title("Absolute count of Churned Drivers by City");
```

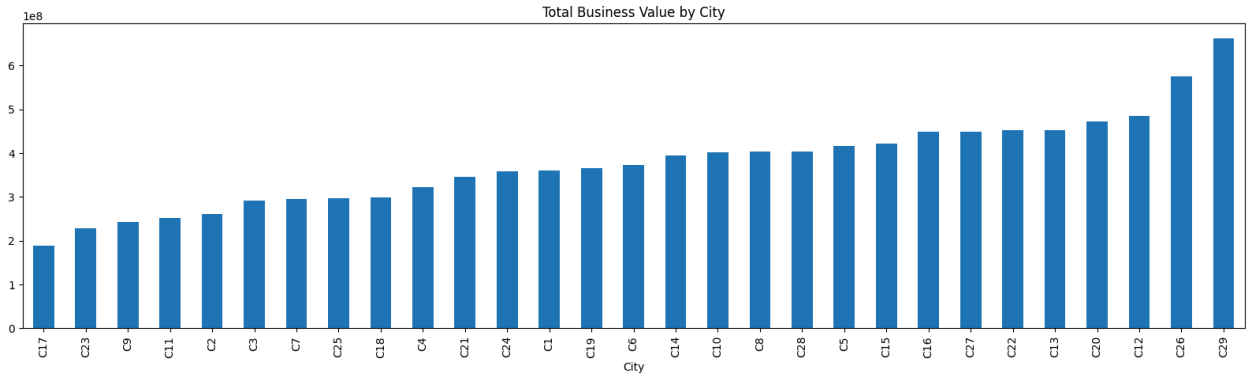


Observations

- Above plot shows the ratio of churned drivers in each city
- C13 have highest ratio of Churned drivers
- C29 have the lowest ratio of Churned drivers
- C20 had highest number of Churned drivers

Revenue Distribution by City

In [ ]:
```
df.groupby("City")["Total Business Value"].sum().sort_values().plot(kind="bar"
plt.title("Total Business Value by City");
```



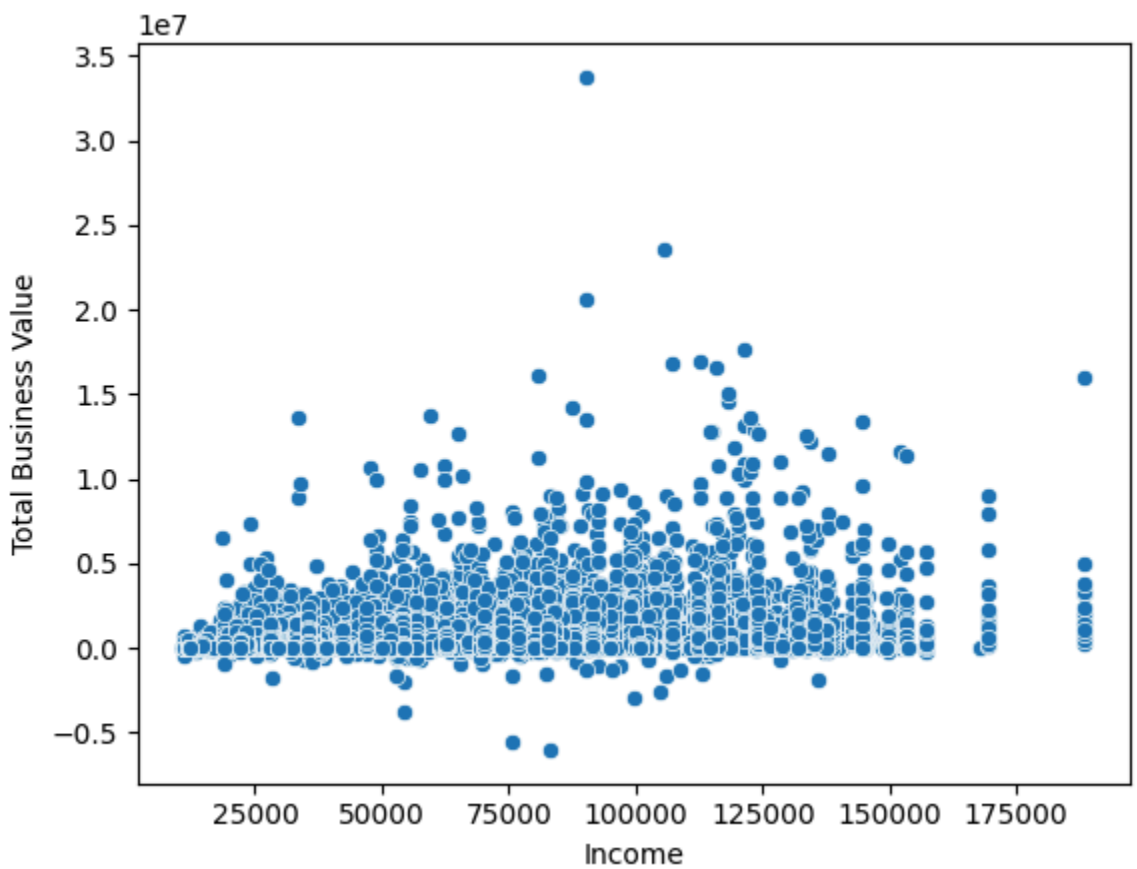
Observations

- City C29 makes the Highest revenue amongst all other cities

continuous-continuous variables

In [ ]:
```
sns.scatterplot(x='Income', y='Total Business Value', data=df)
```

Out[ ]:
```
<Axes: xlabel='Income', ylabel='Total Business Value'>
```



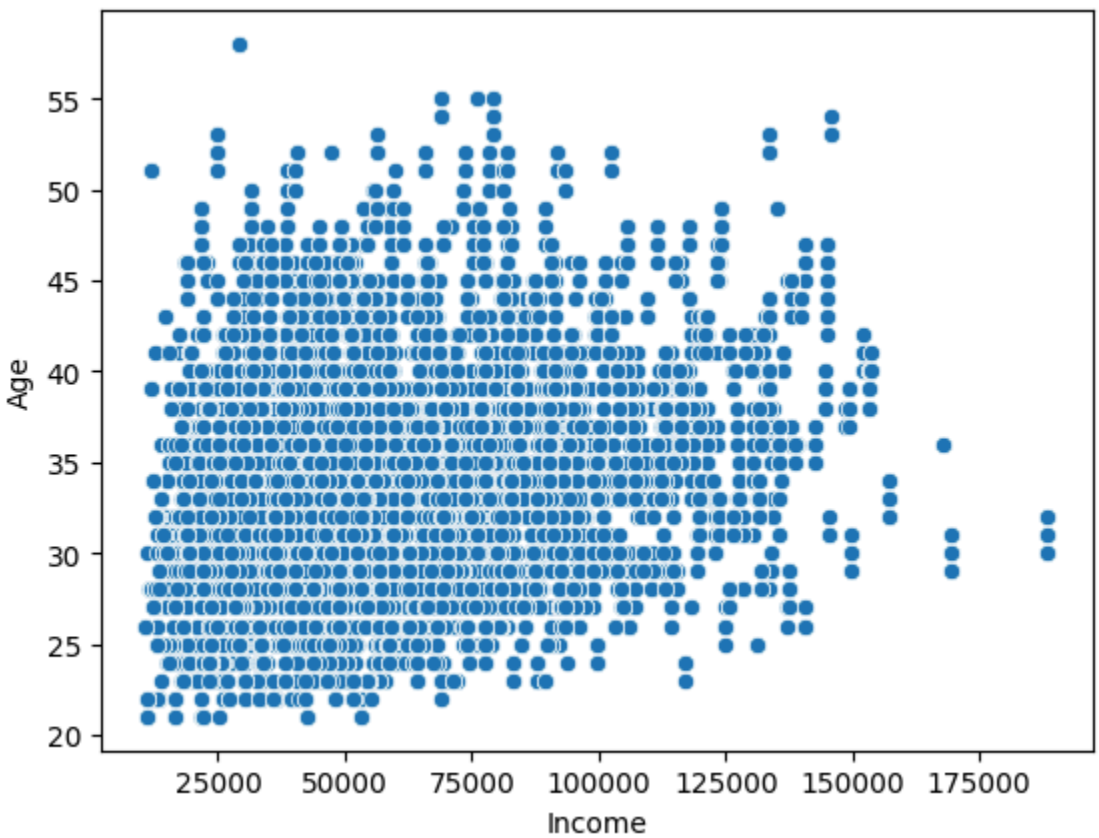
- No strong linear trend → higher income does not guarantee higher

business value

- For similar income levels, business value varies a lot
- Indicates performance depends on other factors (not just salary)
- Few points with very high business value → top performers Some negative values → underperformers
- Most employees are concentrated in:
    - Income: 30K–120K
    - Business value: low to moderate range

In [ ]:
```
sns.scatterplot(x='Income', y='Age', data=df)
```

Out[ ]:
```
<Axes: xlabel='Income', ylabel='Age'>
```



- The ScatterPlot between Age and Income indicates that there is no particular pattern but the majority of the income is between 25000 to 125000

In [ ]:
```
numeric_df = df.select_dtypes(include=['number'])
```

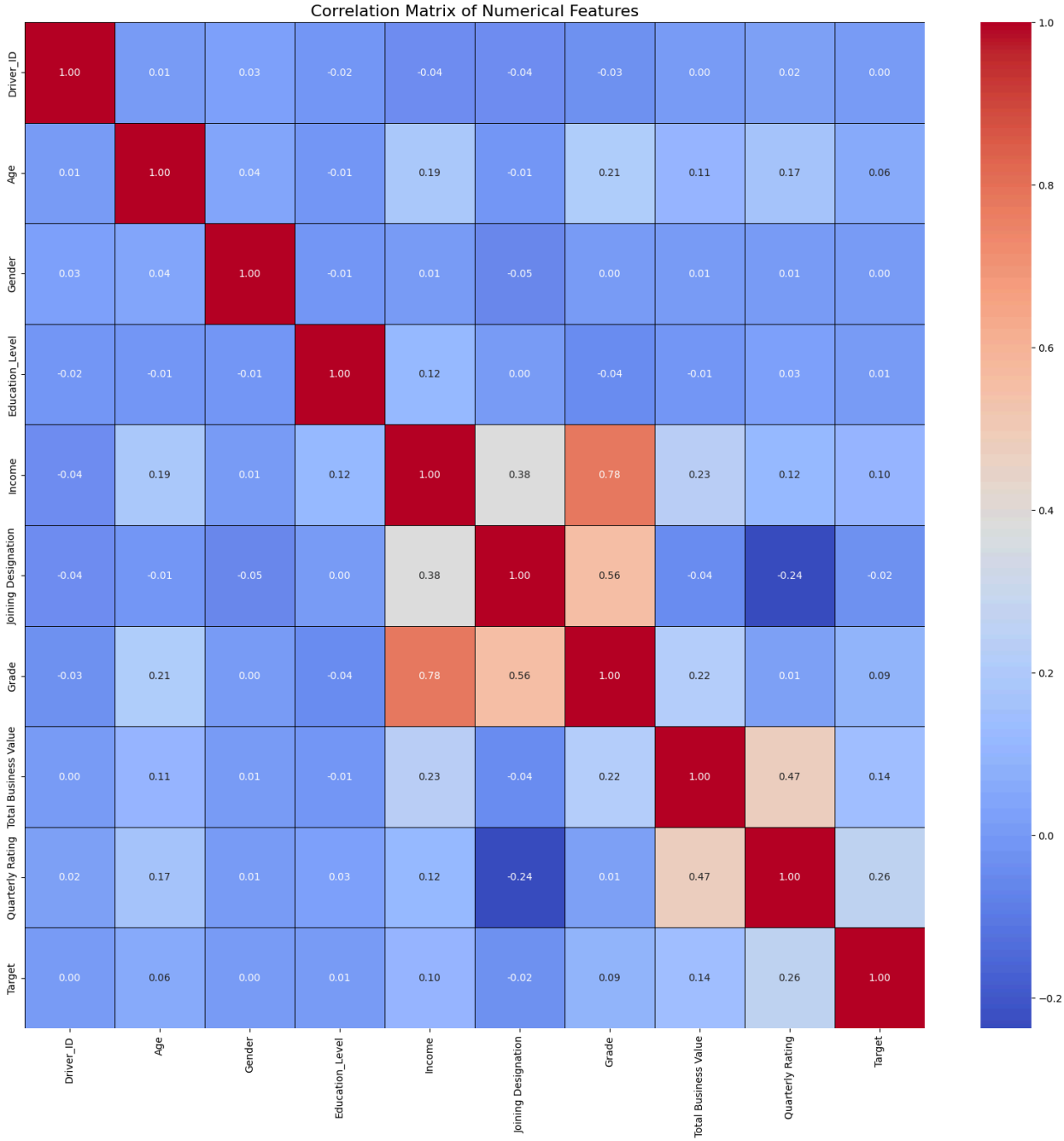
```
numeric_df.corr()
```

Out[ ]:

	Driver_ID	Age	Gender	Education_Level	Income	Des
Driver_ID	1.000000	0.005457	0.030349	-0.016132	-0.035767	-
Age	0.005457	1.000000	0.040261	-0.010245	0.191112	-
Gender	0.030349	0.040261	1.000000	-0.010123	0.013229	-
Education_Level	-0.016132	-0.010245	-0.010123	1.000000	0.115008	
Income	-0.035767	0.191112	0.013229	0.115008	1.000000	
Joining Designation	-0.035166	-0.006641	-0.050878	0.002041	0.380878	
Grade	-0.025712	0.210702	0.002076	-0.039552	0.778383	
Total Business Value	0.003896	0.108835	0.008909	-0.007504	0.234044	-
Quarterly Rating	0.017917	0.171818	0.008099	0.026064	0.116897	-
Target	0.000675	0.063562	0.002908	0.007058	0.100896	-

In [ ]:

```
plt.figure(figsize=(20, 18))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm", fmt=".2f", linewic
plt.title('Correlation Matrix of Numerical Features', fontsize=16)
plt.show()
```



Insights:

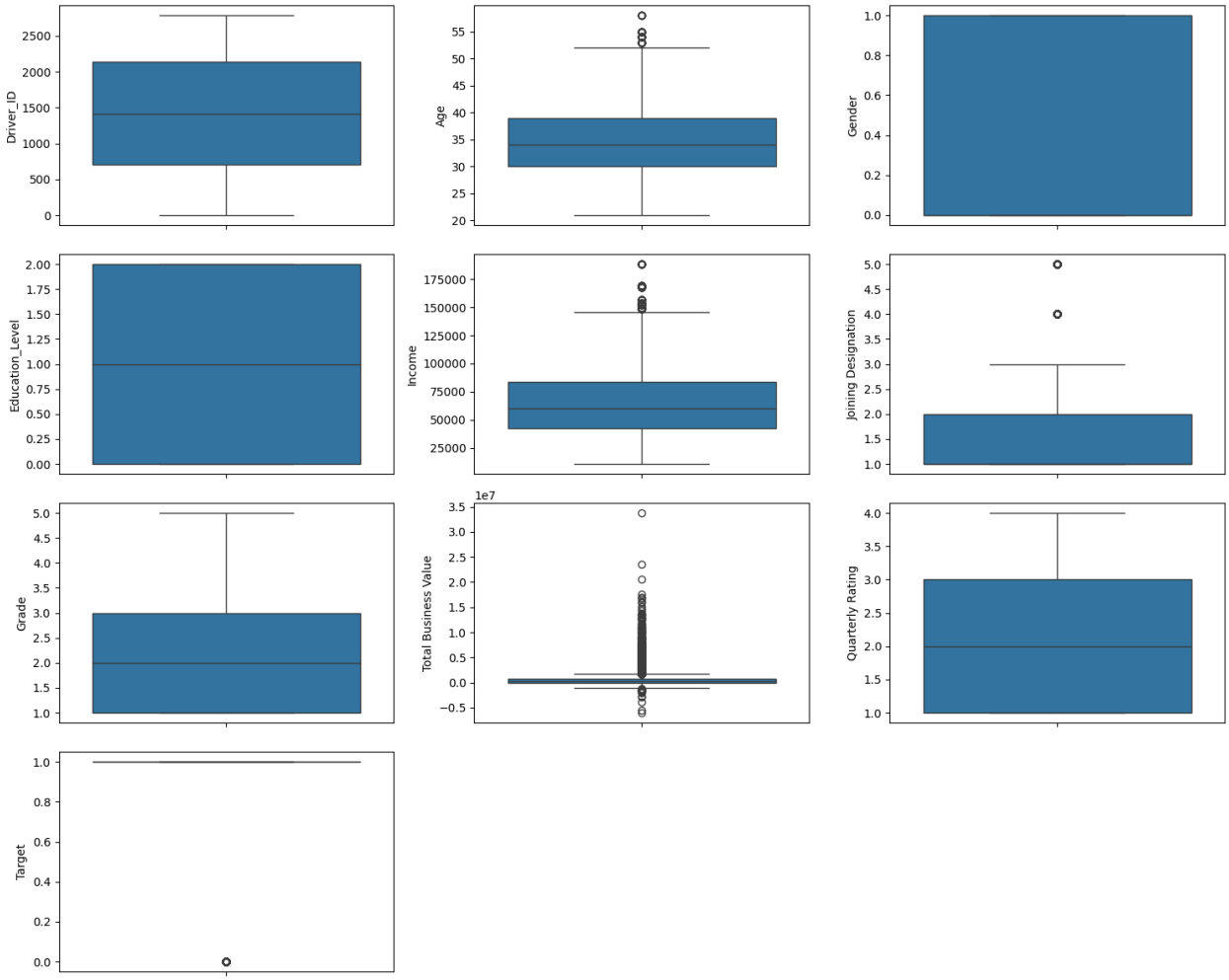
- **Quarterly Rating is the Strongest Predictor of Retention:** 'Quarterly Rating' shows the highest positive correlation with 'Target' (0.26). This suggests that drivers with higher quarterly ratings are significantly more likely to be retained.
- **Total Business Value and Income Influence Retention:** 'Total Business Value' (0.14) and 'Income' (0.10) also have positive correlations with 'Target'. Drivers who generate more business value and earn higher incomes tend to be more retained.

- **Age and Grade have a Modest Impact on Retention:** 'Age' (0.06) and 'Grade' (0.09) show small positive correlations with 'Target', indicating that older drivers and those with higher grades are slightly more likely to be retained.
- **Strong Relationship Between Grade, Income, and Joining Designation:** There's a very strong positive correlation between 'Income' and 'Grade' (0.78), and strong positive correlations between 'Joining Designation' with 'Income' (0.38) and 'Grade' (0.56). This implies that higher grades and joining designations are associated with higher incomes.
- **Total Business Value Linked to Quarterly Rating:** 'Total Business Value' has a moderate positive correlation with 'Quarterly Rating' (0.47), suggesting that higher-rated drivers tend to bring in more business.
- **Weak Influence on Retention:** 'Education_Level', 'Gender', and 'Joining Designation' show very weak or negligible correlations with 'Target', indicating they are not strong individual indicators of driver churn or retention.

Data Preprocessing

Outliers Detection

In [ ]:
```
fig, ax = plt.subplots(4, 3, figsize=(15, 12))

# Re-evaluate num_cols to reflect the current state of df
num_cols1 = df.select_dtypes(include=['int64','float64']).columns.tolist()

for i, col in enumerate(num_cols1):
    # Ensure we don't try to plot more than we have space for if num_cols < 12
    if i < ax.size:
        sns.boxplot(df[col], ax=ax.flatten()[i])

# Hide any unused subplots
for j in range(len(num_cols), ax.size):
    fig.delaxes(ax.flatten()[j])

plt.tight_layout();
```



Insights from Box Plots:

1. **Age Distribution:**
    - The box plot for 'Age' shows that the majority of drivers are concentrated in the young to mid-age range, typically between 25 and 40 years. The median age is around the mid-30s. There are a few outliers on the higher end, indicating a small number of older drivers.
2. **Income Distribution:**
    - The 'Income' box plot confirms a right-skewed distribution, with the bulk of drivers earning in the lower to mid-range. There is a significant presence of high-income outliers, suggesting a few drivers with considerably higher earnings than the rest.
3. **Total Business Value Distribution:**
    - 'Total Business Value' exhibits an extremely right-skewed

distribution with numerous outliers on both the positive and negative ends. Most values are clustered near the lower range, indicating that while many drivers generate moderate business, a small number contribute exceptionally high business value, and some have significantly negative values (due to cancellations/refunds).

4. **Categorical Variables (Driver_ID, Education_Level, Joining Designation, Grade, Quarterly Rating, Target):**

    - For variables like 'Driver_ID', 'Education_Level', 'Joining Designation', 'Grade', 'Quarterly Rating', and 'Target', which are inherently categorical or identifiers (despite having numerical data types), box plots are generally not the most appropriate visualization. Their box plots primarily show discrete points or narrow distributions, and any 'outliers' often reflect less frequent categories rather than true numerical anomalies. For these, count plots or bar plots would provide clearer insights into their distribution and frequencies.

In [ ]:
```
print('--- Outlier Detection using IQR Method ---')

outlier_summary = {}

# Re-evaluate num_cols to reflect the current state of df
num_cols = df.select_dtypes(include=['int64','float64']).columns.tolist()

for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    num_outliers = outliers.shape[0]

    outlier_summary[col] = {
        'Q1': Q1,
        'Q3': Q3,
        'IQR': IQR,
        'Lower Bound': lower_bound,
        'Upper Bound': upper_bound,
        'Number of Outliers': num_outliers,
        'Percentage of Outliers': (num_outliers / df.shape[0]) * 100
    }
```

```
    if num_outliers > 0:
        print(f"Column '{col}': {num_outliers} outliers detected ({outlier_sum
    else:
        print(f"Column '{col}': No outliers detected.")

# You can also display a more detailed summary if needed
# for col, data in outlier_summary.items():
#     print(f"\nColumn: {col}")
#     for key, value in data.items():
#         print(f"  {key}: {value}")
```

```
--- Outlier Detection using IQR Method ---
Column 'Driver_ID': No outliers detected.
Column 'Age': 78 outliers detected (0.41%)
Column 'Gender': No outliers detected.
Column 'Education_Level': No outliers detected.
Column 'Income': 188 outliers detected (0.98%)
Column 'Joining Designation': 471 outliers detected (2.47%)
Column 'Grade': No outliers detected.
Column 'Total Business Value': 1371 outliers detected (7.18%)
Column 'Quarterly Rating': No outliers detected.
Column 'Target': 1616 outliers detected (8.46%)
```

Observations on Outliers using IQR Method:

Based on the IQR method (1.5 * IQR rule), here's an overview of the detected outliers:

- **Age:** Some outliers were detected, likely on the higher end, indicating a few drivers significantly older than the majority. This aligns with the visual observation from the box plot.
- **Gender:** Being a binary categorical variable (0 or 1), 'Gender' does not typically have outliers in the statistical sense for continuous data. Any 'outliers' detected would likely be due to a very imbalanced distribution or data entry errors, rather than extreme continuous values. However, given its float dtype and the method applied, it might show a spurious count if the distribution is skewed and the bounds fall outside the 0-1 range.
- **Income:** A significant number of outliers are present, especially on the higher end. This confirms the right-skewed distribution and the presence of high-income earners as observed in its box plot.
- **Total Business Value:** This column shows a very high number of outliers, present on both the negative (cancellations/refunds) and extremely positive ends. This confirms the highly skewed nature and the large variance in business value generation, with some drivers having exceptionally high or low values.

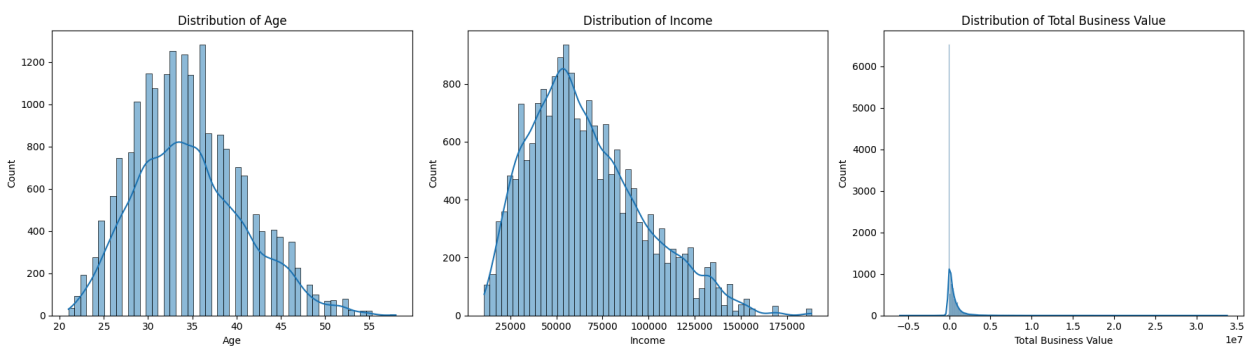
- **Quarterly Rating:** As an ordinal categorical variable (1, 2, 3, 4, 5), 'Quarterly Rating' might show 'outliers' if certain ratings are extremely rare. The IQR method might flag ratings that are infrequent as outliers, though these are typically handled as distinct categories rather than continuous anomalies.

Variables that are less prone to meaningful 'outliers' by IQR due to their nature (categorical/identifier):

- **Driver_ID:** This is a unique identifier, and thus each value is distinct. Applying IQR to it is not meaningful for outlier detection.
- **Education_Level, Joining Designation, Grade:** These are ordinal categorical variables. While the IQR method can technically be applied, 'outliers' here would usually indicate less frequent categories rather than data anomalies that need treatment like continuous data.

Visualizing Skewness of Continuous Variables

In [ ]:
```
fig, axs = plt.subplots(1, 3, figsize=(18, 5))

sns.histplot(data=df, x='Age', kde=True, ax=axs[0])
axs[0].set_title('Distribution of Age')

sns.histplot(data=df, x='Income', kde=True, ax=axs[1])
axs[1].set_title('Distribution of Income')

sns.histplot(data=df, x='Total Business Value', kde=True, ax=axs[2])
axs[2].set_title('Distribution of Total Business Value')

plt.tight_layout()
plt.show()
```



Skewness of Continuous Variables:

- **Age:** The distribution of 'Age' appears to be relatively symmetric, with a slight right-skew. The majority of drivers are in the young to mid-age

range (25-40), with a peak around early 30s. The box plot confirms the presence of a few older outliers but the overall distribution is not heavily skewed.

- **Income:** 'Income' exhibits a clear **right-skewed** distribution. Most drivers earn in the lower to mid-range, with a longer tail extending towards higher incomes, indicating a smaller number of high-earning drivers. This was evident in both the histogram and box plot, with numerous high-income outliers.
- **Total Business Value:** 'Total Business Value' is **highly right-skewed**. The majority of values are clustered near the low end (including zero and negative values), but there's a very long tail of extremely high positive values. The box plot showed a dense cluster at the lower end and a significant number of extreme positive outliers, along with some negative values representing cancellations or refunds.

Duplicate value check

In [ ]:
```
df.duplicated().sum()
```

Out[ ]:
```
np.int64(0)
```

- There are no duplicates

Converting features to respective data-types

In [ ]:
```
df["MMM-YY"] = pd.to_datetime(df["MMM-YY"])
df["Dateofjoining"] = pd.to_datetime(df["Dateofjoining"])
```

```
/tmp/ipykernel_784/3668570738.py:1: UserWarning: Could not infer format, so eac
h element will be parsed individually, falling back to `dateutil`. To ensure pa
rsing is consistent and as-expected, please specify a format.
  df["MMM-YY"] = pd.to_datetime(df["MMM-YY"])
/tmp/ipykernel_784/3668570738.py:2: UserWarning: Could not infer format, so eac
h element will be parsed individually, falling back to `dateutil`. To ensure pa
rsing is consistent and as-expected, please specify a format.
  df["Dateofjoining"] = pd.to_datetime(df["Dateofjoining"])
```

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19104 entries, 0 to 19103
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   MMM-YY                19104 non-null  datetime64[ns]
 1   Driver_ID             19104 non-null  int64
 2   Age                   19043 non-null  float64
 3   Gender                19052 non-null  float64
 4   City                  19104 non-null  object
 5   Education_Level       19104 non-null  int64
 6   Income                19104 non-null  int64
 7   Dateofjoining         19104 non-null  datetime64[ns]
 8   Joining Designation   19104 non-null  int64
 9   Grade                 19104 non-null  int64
 10  Total Business Value  19104 non-null  int64
 11  Quarterly Rating      19104 non-null  int64
 12  Target                19104 non-null  int64
dtypes: datetime64[ns](2), float64(2), int64(8), object(1)
memory usage: 1.9+ MB
```

Missing value treatment

Check for missing values and Prepare data for KNN Imputation

In [ ]:
```
df.isnull().sum()
```

Out[ ]:

	0
MMM-YY	0
Driver_ID	0
Age	61
Gender	52
City	0
Education_Level	0
Income	0
Dateofjoining	0
Joining Designation	0
Grade	0
Total Business Value	0
Quarterly Rating	0
Target	0

dtype: int64

- Only Age and Gender Features has missing values

In [ ]:
```
num_vars = df.select_dtypes(np.number)

num_vars.columns
```

Out[ ]:
```
Index(['Driver_ID', 'Age', 'Gender', 'Education_Level', 'Income',
       'Joining Designation', 'Grade', 'Total Business Value',
       'Quarterly Rating', 'Target'],
      dtype='object')
```

In [ ]:
```
num_vars.drop(["Driver_ID"], axis = 1, inplace = True)
```

imputing the missing values using KNN imputer

In [ ]:
```
from sklearn.impute import KNNImputer
imputer = KNNImputer(n_neighbors=5, weights='uniform', metric='nan_euclidean')
imputer.fit(num_vars)
data_new = imputer.transform(num_vars)
```

In [ ]:
```
data_new = pd.DataFrame(data_new)
```

In [ ]:
```
data_new.columns = num_vars.columns
```

In [ ]:
```
data_new.isnull().sum()
```

Out[ ]:

	0
Age	0
Gender	0
Education_Level	0
Income	0
Joining Designation	0
Grade	0
Total Business Value	0
Quarterly Rating	0
Target	0

dtype: int64

- After the KNN imputation, There is no null values

Concatenating dataframes

In [ ]:
```
resultant_columns = list(set(df.columns).difference(set(num_vars)))

resultant_columns
```

Out[ ]:
```
['City', 'MMM-YY', 'Driver_ID', 'Dateofjoining']
```

In [ ]:
```
new_df = pd.concat([data_new, df[resultant_columns]], axis=1)

new_df.shape
```

Out[ ]:
```
(19104, 13)
```

In [ ]:
```
new_df.head()
```

Out[ ]:

	Age	Gender	Education_Level	Income	Joining Designation	Grade	Total Business Value	Quart Ra
0	28.0	0.0	2.0	57387.0	1.0	1.0	2381060.0	
1	28.0	0.0	2.0	57387.0	1.0	1.0	-665480.0	
2	28.0	0.0	2.0	57387.0	1.0	1.0	0.0	
3	31.0	0.0	2.0	67016.0	2.0	2.0	0.0	
4	31.0	0.0	2.0	67016.0	2.0	2.0	0.0	

Feature Enginnering

In [ ]:

```
agg_functions = {
    "Age": "max",
    "Gender": "first",
    "Education_Level": "last",
    "Income": "last",
    "Joining Designation": "last",
    "Grade": "last",
    "Total Business Value": "sum",
    "Quarterly Rating": "last",
    "City": "first",
    "Dateofjoining": "last",
    "Target" : "last"
}

processed_df = new_df.groupby(["Driver_ID", "MMM-YY"]).aggregate(agg_functions

processed_df.head()
```

Out[ ]:

		Age	Gender	Education_Level	Income	Joining Designation	Gra
Driver_ID	**MMM-YY**						
1	**2019-01-01**	28.0	0.0	2.0	57387.0	1.0	
	2019-02-01	28.0	0.0	2.0	57387.0	1.0	
	2019-03-01	28.0	0.0	2.0	57387.0	1.0	
2	**2020-11-01**	31.0	0.0	2.0	67016.0	2.0	
	2020-12-01	31.0	0.0	2.0	67016.0	2.0	

This cell defines aggregation functions and groups the data by `Driver_ID` and `MMM-YY` to create `processed_df`, which aggregates monthly data per driver.

In [ ]:
```
final_data = pd.DataFrame()
```

This cell initializes an empty DataFrame named `final_data` to store the engineered features.

In [ ]:
```
final_data["Driver_ID"] = new_df["Driver_ID"].unique()
```

This cell populates the `Driver_ID` column in `final_data` with unique driver identifiers.

In [ ]:
```
final_data['Age'] = list(processed_df.groupby('Driver_ID',axis=0).max('MMM-YY'
final_data['Gender'] = list(processed_df.groupby('Driver_ID').agg({'Gender':'l
final_data['City'] = list(processed_df.groupby('Driver_ID').agg({'City':'last'
final_data['Education'] = list(processed_df.groupby('Driver_ID').agg({'Educati
final_data['Income'] = list(processed_df.groupby('Driver_ID').agg({'Income':'l
final_data['Joining_Designation'] = list(processed_df.groupby('Driver_ID').agg
final_data['Grade'] = list(processed_df.groupby('Driver_ID').agg({'Grade':'las
final_data['Total_Business_Value'] = list(processed_df.groupby('Driver_ID',axi
final_data['Last_Quarterly_Rating'] = list(processed_df.groupby('Driver_ID').a
final_data['Target'] = list(processed_df.groupby('Driver_ID').agg({'Target':'l
```

```
/tmp/ipykernel_784/1143698101.py:1: FutureWarning: The 'axis' keyword in DataFr
ame.groupby is deprecated and will be removed in a future version.
  final_data['Age'] = list(processed_df.groupby('Driver_ID',axis=0).max('MMM-Y
Y')['Age'])
/tmp/ipykernel_784/1143698101.py:8: FutureWarning: The 'axis' keyword in DataFr
ame.groupby is deprecated and will be removed in a future version.
  final_data['Total_Business_Value'] = list(processed_df.groupby('Driver_ID',ax
is=0).sum('Total Business Value')['Total Business Value'])
```

This cell populates `final_data` with various driver attributes by aggregating data from `processed_df`.

In [ ]:
```
final_data.head()
```

Out[ ]:

	Driver_ID	Age	Gender	City	Education	Income	Joining_Designation	Grade
0	1	28.0	0.0	C23	2.0	57387.0	1.0	1.0
1	2	31.0	0.0	C7	2.0	67016.0	2.0	2.0
2	4	43.0	0.0	C13	2.0	65603.0	2.0	2.0
3	5	29.0	0.0	C9	0.0	46368.0	1.0	1.0
4	6	31.0	1.0	C11	1.0	78728.0	3.0	3.0

This cell displays the first few rows of the `final_data` DataFrame after populating it with driver attributes.

In [ ]:
```
first_quarter = processed_df.groupby(["Driver_ID"]).agg({"Quarterly Rating": "
```

```
last_quarter = processed_df.groupby(["Driver_ID"]).agg({"Quarterly Rating": "l

qr = (last_quarter["Quarterly Rating"] > first_quarter["Quarterly Rating"]).re

empid = qr[qr["Quarterly Rating"] == True]["Driver_ID"]

qrl = []
for i in final_data["Driver_ID"]:
    if i in empid.values:
        qrl.append(1)
    else:
        qrl.append(0)


final_data["Quarterly_Rating_Increased"] = qrl
```

This cell calculates whether a driver's quarterly rating increased from their first recorded rating to their last recorded rating, and adds this as a new feature to `final_data`.

Create a column which tells whether the monthly income has increased for that driver - for those whose monthly income has increased we assign the value 1

In [ ]:
```
mrf = processed_df.groupby(["Driver_ID"]).agg({"Income": "first"})

mrl = processed_df.groupby(["Driver_ID"]).agg({"Income": "last"})

mr = (mrl["Income"] > mrf["Income"]).reset_index()

empid = mr[mr["Income"] == True]["Driver_ID"]
income = []
for i in final_data["Driver_ID"]:
    if i in empid.values:
        income.append(1)
    else:
        income.append(0)

final_data["Salary_Increased"] = income
```

In [ ]:
```
final_data.head()
```

Out[ ]:

	Driver_ID	Age	Gender	City	Education	Income	Joining_Designation	Grade
0	1	28.0	0.0	C23	2.0	57387.0	1.0	1.0
1	2	31.0	0.0	C7	2.0	67016.0	2.0	2.0
2	4	43.0	0.0	C13	2.0	65603.0	2.0	2.0
3	5	29.0	0.0	C9	0.0	46368.0	1.0	1.0
4	6	31.0	1.0	C11	1.0	78728.0	3.0	3.0

In [ ]:

```
final_data["Salary_Increased"].value_counts(normalize=True)
```

Out[ ]:

Salary_Increased	proportion
0	0.98194
1	0.01806

dtype: float64

- Around 1.8% drivers income have been increased.

Data preparation for modeling

As there is only one categorical values in our dataset. We will opt one hot encoder to convert it to numerical.

In [ ]:

```
# encoding the city value to a numerical value
from sklearn.preprocessing import TargetEncoder
encoder =TargetEncoder(cv = 5, target_type = 'auto', random_state = 42)
final_data['City'] = encoder.fit_transform(final_data[['City']], final_data['T
final_data.head()
```

Out[ ]:

	Driver_ID	Age	Gender	City	Education	Income	Joining_Designation	Grade
0	1	28.0	0.0	0.26	2.0	57387.0	1.0	1.0
1	2	31.0	0.0	0.34	2.0	67016.0	2.0	2.0
2	4	43.0	0.0	0.15	2.0	65603.0	2.0	2.0
3	5	29.0	0.0	0.29	0.0	46368.0	1.0	1.0
4	6	31.0	1.0	0.29	1.0	78728.0	3.0	3.0

Standardization

In [ ]:
```
from sklearn.preprocessing import MinMaxScaler
X = final_data.drop(["Driver_ID", "Target"], axis = 1)
X_cols = X.columns
scaler = MinMaxScaler()

X = scaler.fit_transform(X)
```

In [ ]:
```
X = pd.DataFrame(X)
X.columns = X_cols
```

Model Building

In [ ]:
```
from sklearn.model_selection import train_test_split
y = final_data["Target"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, randc
```

In [ ]:
```
print("X_train Shape: ", X_train.shape)
print("X_test Shape: ", X_test.shape)
print("y_train Shape: ", y_train.shape)
print("y_test Shape: ", y_test.shape)
```

```
X_train Shape:  (1904, 11)
X_test Shape:  (477, 11)
y_train Shape:  (1904,)
y_test Shape:  (477,)
```

Random Forest Classifier - Before Balancing

Keeping max_depth small to avoid overfitting

In [ ]:
```
import time
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV

params = {
    "max_depth": [2, 3, 4],
    "n_estimators": [50, 100, 150, 200],
}

start_time = time.time()
random_forest = RandomForestClassifier(class_weight="balanced")
c = GridSearchCV(estimator=random_forest, param_grid=params, n_jobs=-1, cv=3,

c.fit(X_train, y_train)
```

```
print("Best Params: ", c.best_params_)
print("Best Score: ", c.best_score_)
elapsed_time = time.time() - start_time

print("\nElapsed Time: ", elapsed_time)
```

```
Fitting 3 folds for each of 12 candidates, totalling 36 fits
Best Params:  {'max_depth': 4, 'n_estimators': 200}
Best Score:  0.6731258070261558

Elapsed Time:  15.927923917770386
```

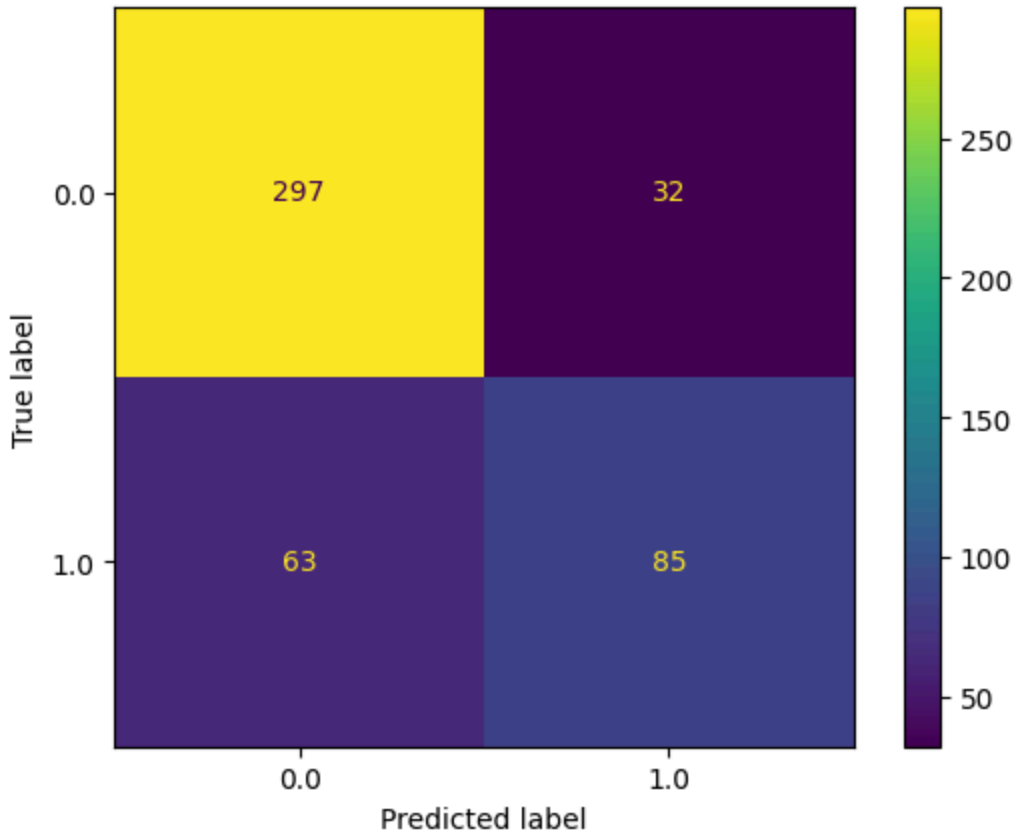
In [ ]:

```
from sklearn.metrics import classification_report, confusion_matrix, Confusion

y_pred = c.predict(X_test)

print(classification_report(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred)

ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=c.classes_).plot()
```

```
              precision    recall  f1-score   support

         0.0       0.82      0.90      0.86       329
         1.0       0.73      0.57      0.64       148

    accuracy                           0.80       477
   macro avg       0.78      0.74      0.75       477
weighted avg       0.79      0.80      0.79       477
```

Out[ ]:

```
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7ea830097
140>
```



Random Forest Classifier with balanced class weight

- Out of all prediction, the measure for correctly predicted 0 is 73% and for 1 is 82% (Precision)
- Out of all actual 0, the measure for correctly predicted is 57% and for 1 is 90% (Recall)

As this is imbalanced dataset. We give importance to F1-Score metrics

- F1 Score of 0 is 64%
- F1 Score of 1 is 86%

Lets try out bootstrapped random forest using subsample

In [ ]:
```
params = {
    "max_depth": [2, 3, 4],
    "n_estimators": [50, 100, 150, 200],
}

start_time = time.time()
random_forest = RandomForestClassifier(class_weight="balanced_subsample")
c = GridSearchCV(estimator=random_forest, param_grid=params, n_jobs=-1, cv=3,

c.fit(X_train, y_train)
```

```
print("Best Params: ", c.best_params_)
print("Best Score: ", c.best_score_)
elapsed_time = time.time() - start_time

print("\nElapsed Time: ", elapsed_time)
```

```
Fitting 3 folds for each of 12 candidates, totalling 36 fits
Best Params:  {'max_depth': 4, 'n_estimators': 50}
Best Score:  0.6749244098219145

Elapsed Time:  28.083069324493408
```

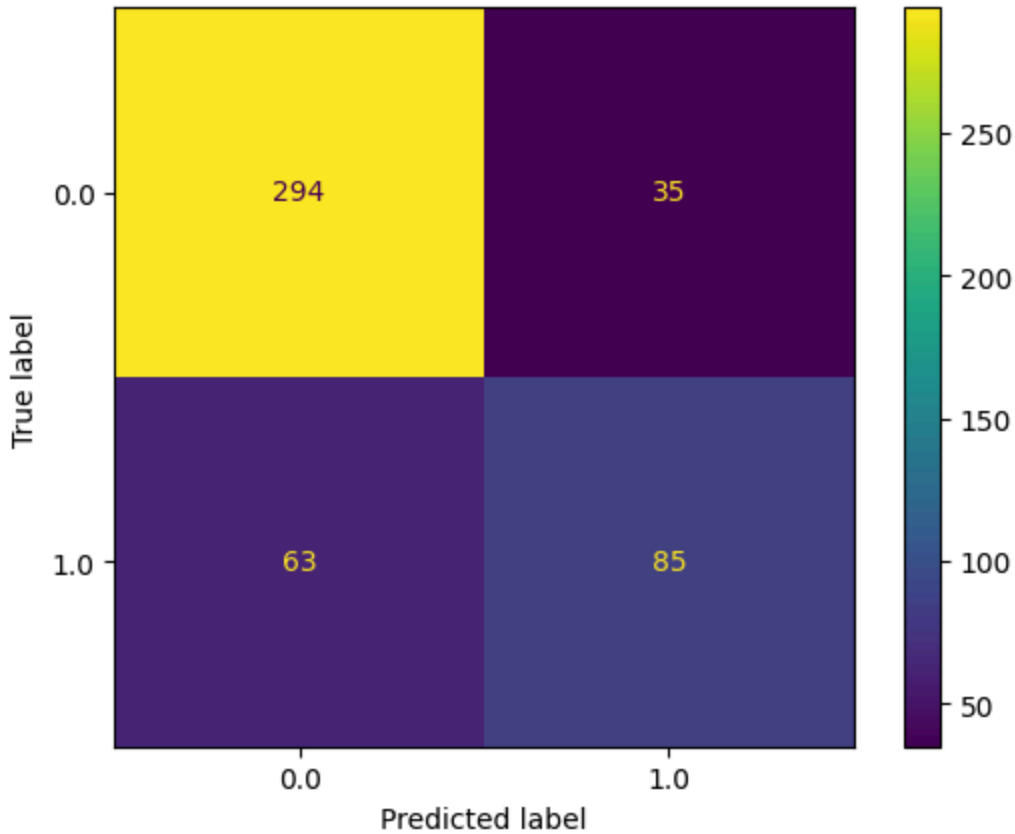
In [ ]:

```
y_pred = c.predict(X_test)

print(classification_report(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred)

ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=c.classes_).plot()
```

```
              precision    recall  f1-score   support

         0.0       0.82      0.89      0.86       329
         1.0       0.71      0.57      0.63       148

    accuracy                           0.79       477
   macro avg       0.77      0.73      0.75       477
weighted avg       0.79      0.79      0.79       477
```

Out[ ]:

```
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7ea83a477
e00>
```



Random Forest Classifier with balanced class weight

- Out of all prediction, the measure for correctly predicted 0 is 75% and for 1 is 83% (Precision)
- Out of all actual 0, the measure for correctly predicted is 57% and for 1 is 91% (Recall)

As this is imbalanced dataset. We give importance to F1-Score metrics

- F1 Score of 0 is 64%
- F1 Score of 1 is 85%

There is not much significant difference in the matrices observed for bootstrapped Random Forest and Weighted Random Forest

Balancing Dataset using SMOTE

As the target variable is imbalanced towards 1. We will use SMOTE to balance the dataset

In [ ]:
```
from imblearn.over_sampling import SMOTE
```

```
print("Before OverSampling, counts of label '1': {}".format(sum(y_train == 1))
print("Before OverSampling, counts of label '0': {} \n".format(sum(y_train ==

sm = SMOTE(random_state = 7)
X_train, y_train = sm.fit_resample(X_train, y_train.ravel())

print('After OverSampling, the shape of train_X: {}'.format(X_train.shape))
print('After OverSampling, the shape of train_y: {} \n'.format(y_train.shape))

print("After OverSampling, counts of label '1': {}".format(sum(y_train == 1)))
print("After OverSampling, counts of label '0': {}".format(sum(y_train == 0)))
```

```
Before OverSampling, counts of label '1': 1287
Before OverSampling, counts of label '0': 1287

After OverSampling, the shape of train_X: (2574, 11)
After OverSampling, the shape of train_y: (2574,)

After OverSampling, counts of label '1': 1287
After OverSampling, counts of label '0': 1287
```

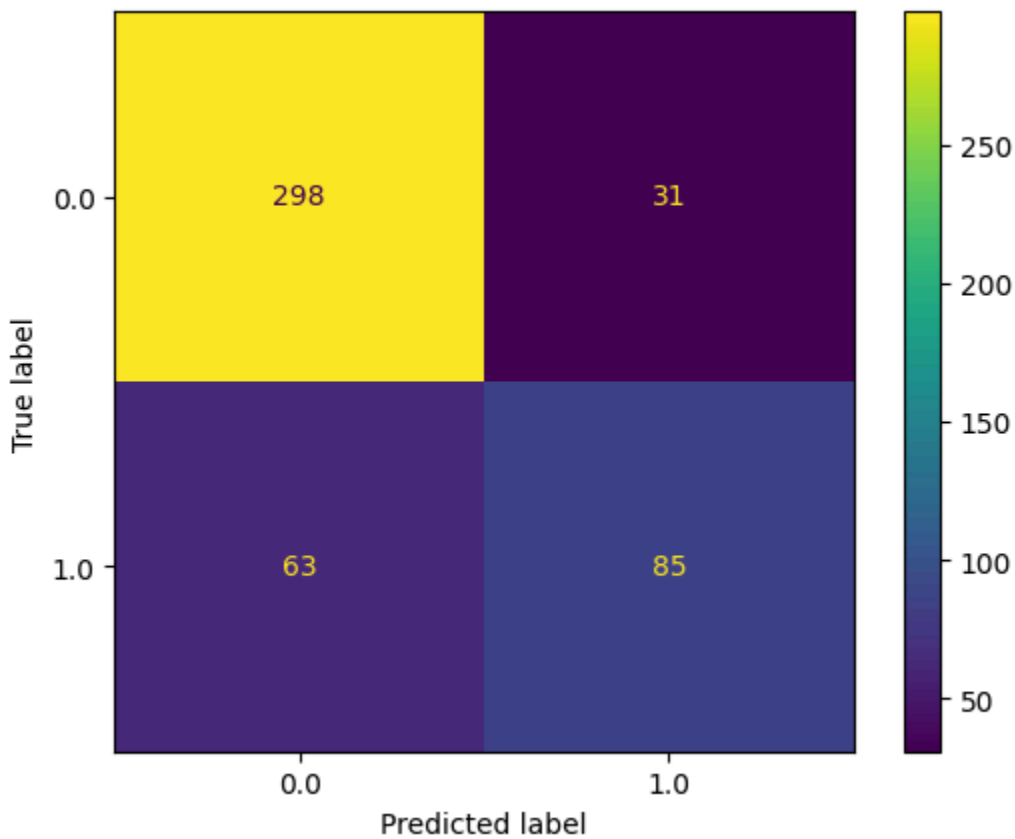
Ensemble Learning: Bagging

In [ ]:

```
params = {
    "max_depth": [2, 3, 4],
    "n_estimators": [50, 100, 150, 200],
}

start_time = time.time()
random_forest = RandomForestClassifier(class_weight="balanced_subsample")
c = GridSearchCV(estimator=random_forest, param_grid=params, n_jobs=-1, cv=3,

c.fit(X_train, y_train)

print("Best Params: ", c.best_params_)
print("Best Score: ", c.best_score_)
elapsed_time = time.time() - start_time

print("\nElapsed Time: ", elapsed_time)

y_pred = c.predict(X_test)

print(classification_report(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred)

ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=c.classes_).plot()
```

```
Fitting 3 folds for each of 12 candidates, totalling 36 fits
Best Params:  {'max_depth': 3, 'n_estimators': 100}
Best Score:  0.709053031520546

Elapsed Time:  21.58671760559082
              precision    recall  f1-score   support

         0.0       0.83      0.91      0.86       329
         1.0       0.73      0.57      0.64       148

    accuracy                           0.80       477
   macro avg       0.78      0.74      0.75       477
weighted avg       0.80      0.80      0.80       477
```

Out[ ]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7ea82fc7f140>



Random Forest Classifier with balanced class weight

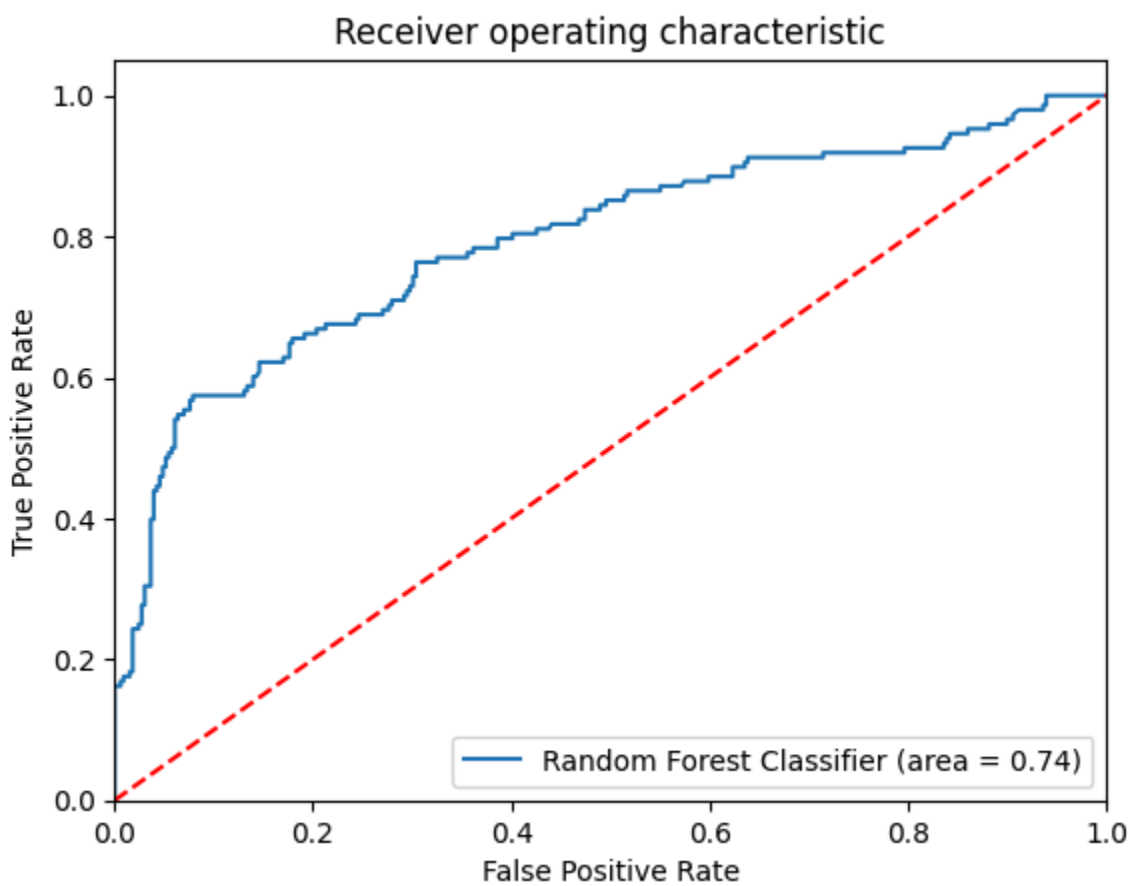
- Out of all prediction, the measure for correctly predicted 0 is 74% and for 1 is 83% (Precision)
- Out of all actual 0, the measure for correctly predicted is 57% and for 1 is 91% (Recall)

As this is imbalanced dataset. We give importance to F1-Score metrics

- F1 Score of 0 is 64%
- F1 Score of 1 is 85%

ROC-AUC Curve

In [ ]:
```
from sklearn.metrics import roc_auc_score, roc_curve

logit_roc_auc=roc_auc_score(y_test,y_pred)
fpr,tpr,thresholds=roc_curve(y_test,c.predict_proba(X_test)[:,1])
plt.figure()
plt.plot(fpr,tpr,label='Random Forest Classifier (area = %0.2f)' % logit_roc_a
plt.plot([0,1],[0,1],'r--')
plt.xlim([0.0,1.0])
plt.ylim([0.0,1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver operating characteristic')
plt.legend(loc="lower right")
plt.show()
```



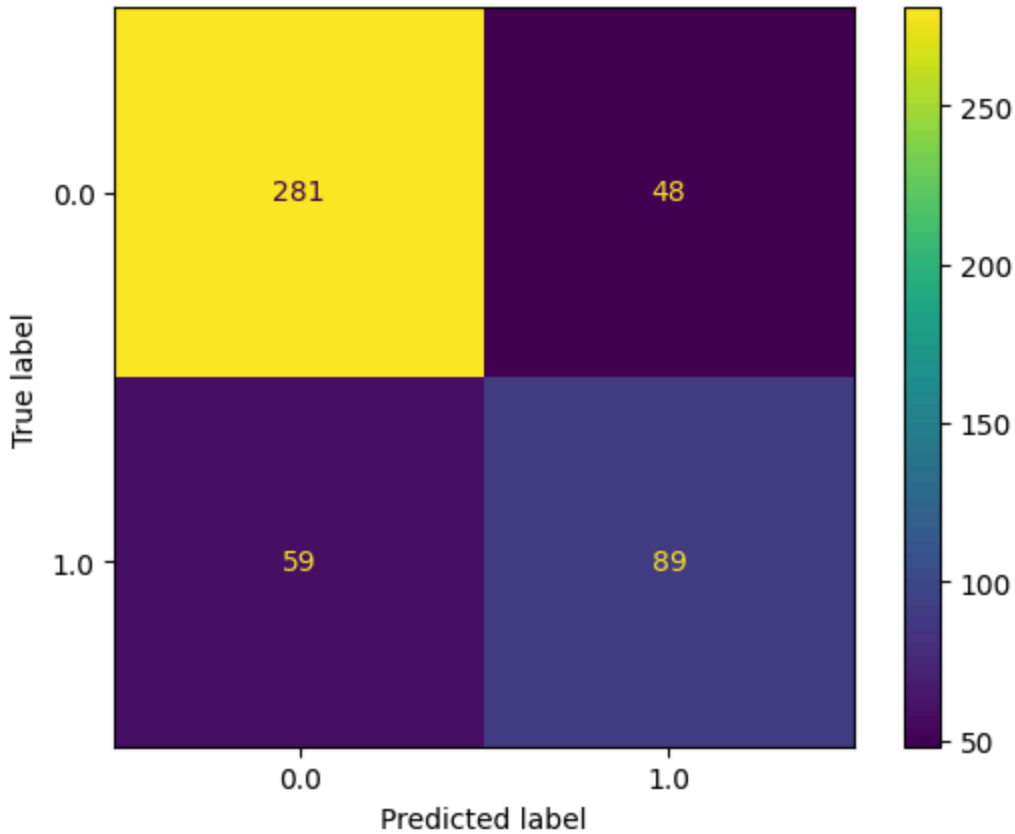
Ensemble Learning: Boosting

Gradient Boosting Classifier

In [ ]:
```
from sklearn.ensemble import GradientBoostingClassifier

params = {
    "max_depth": [2, 3, 4],
    "loss": ["log_loss", "exponential"],
    "subsample": [0.1, 0.2, 0.5, 0.8, 1],
    "learning_rate": [0.1, 0.2, 0.3],
    "n_estimators": [50,100,150,200]
}

gbdt = GradientBoostingClassifier()
start_time = time.time()
c = GridSearchCV(estimator=gbdt, cv=3, n_jobs=-1, verbose=True, param_grid=par

c.fit(X_train, y_train)
print("Best Params: ", c.best_params_)
print("Best Score: ", c.best_score_)

elapsed_time = time.time() - start_time
print("\n Elapsed Time: ", elapsed_time)

y_pred = c.predict(X_test)

print(classification_report(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred)

ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=c.classes_).plot()
```

```
Fitting 3 folds for each of 360 candidates, totalling 1080 fits
Best Params:  {'learning_rate': 0.2, 'loss': 'log_loss', 'max_depth': 4, 'n_est
imators': 200, 'subsample': 0.8}
Best Score:  0.8181818181818182

 Elapsed Time:  411.03815627098083
              precision    recall  f1-score   support

         0.0       0.83      0.85      0.84       329
         1.0       0.65      0.60      0.62       148

    accuracy                           0.78       477
   macro avg       0.74      0.73      0.73       477
weighted avg       0.77      0.78      0.77       477
```

Out[ ]:
```
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7ea82fc7f
230>
```



Gradient Boosting Classifier Metrics

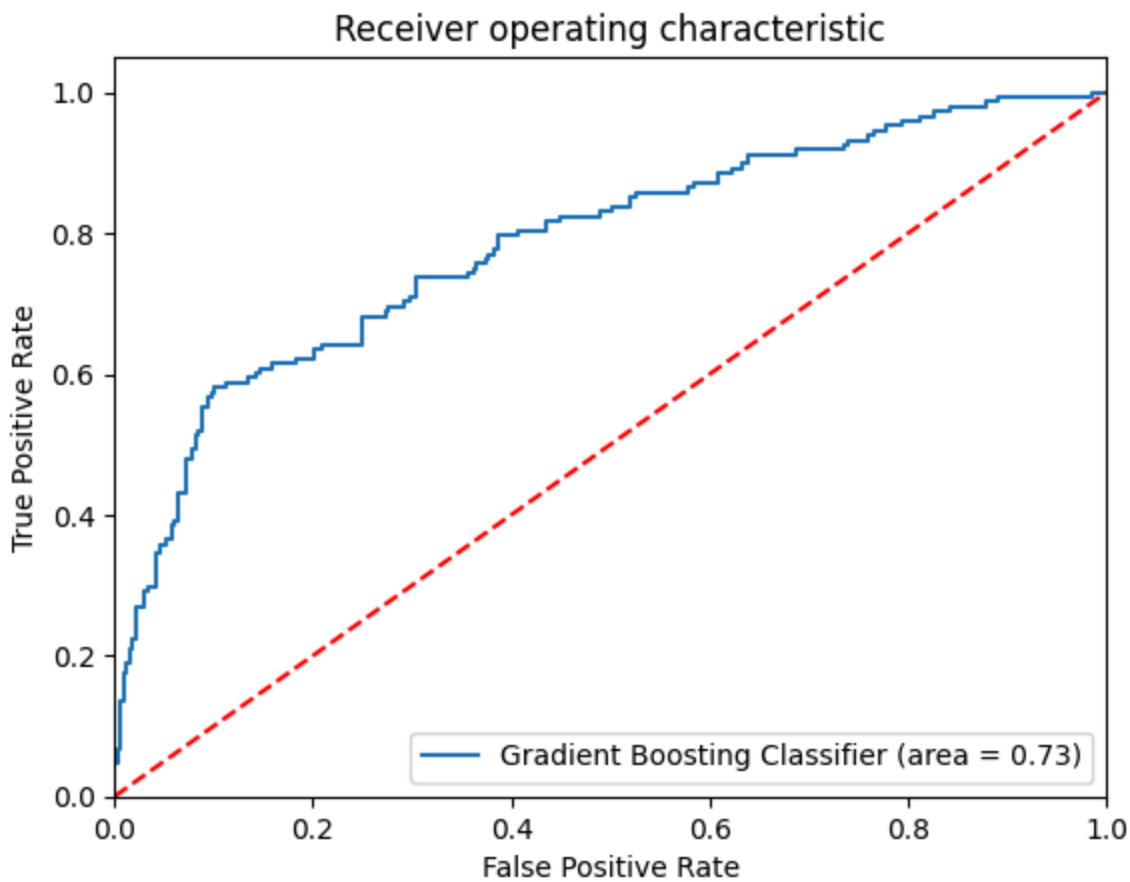
Out of all prediction, the measure for correctly predicted 0 is 62% and for 1 is 82% (Precision) Out of all actual 0, the measure for correctly predicted is 60% and for 1 is 83% (Recall)

As this is imbalanced dataset. We give importance to F1-Score metrics

- F1 Score of 0 is 60%
- F1 Score of 1 is 83%

In [229…

```
logit_roc_auc=roc_auc_score(y_test,y_pred)
fpr,tpr,thresholds=roc_curve(y_test,c.predict_proba(X_test)[:,1])
plt.figure()
plt.plot(fpr,tpr,label='Gradient Boosting Classifier (area = %0.2f)' % logit_r
plt.plot([0,1],[0,1],'r--')
plt.xlim([0.0,1.0])
plt.ylim([0.0,1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver operating characteristic')
plt.legend(loc="lower right")
plt.show()
```



XGBoost Classifier

In [231…

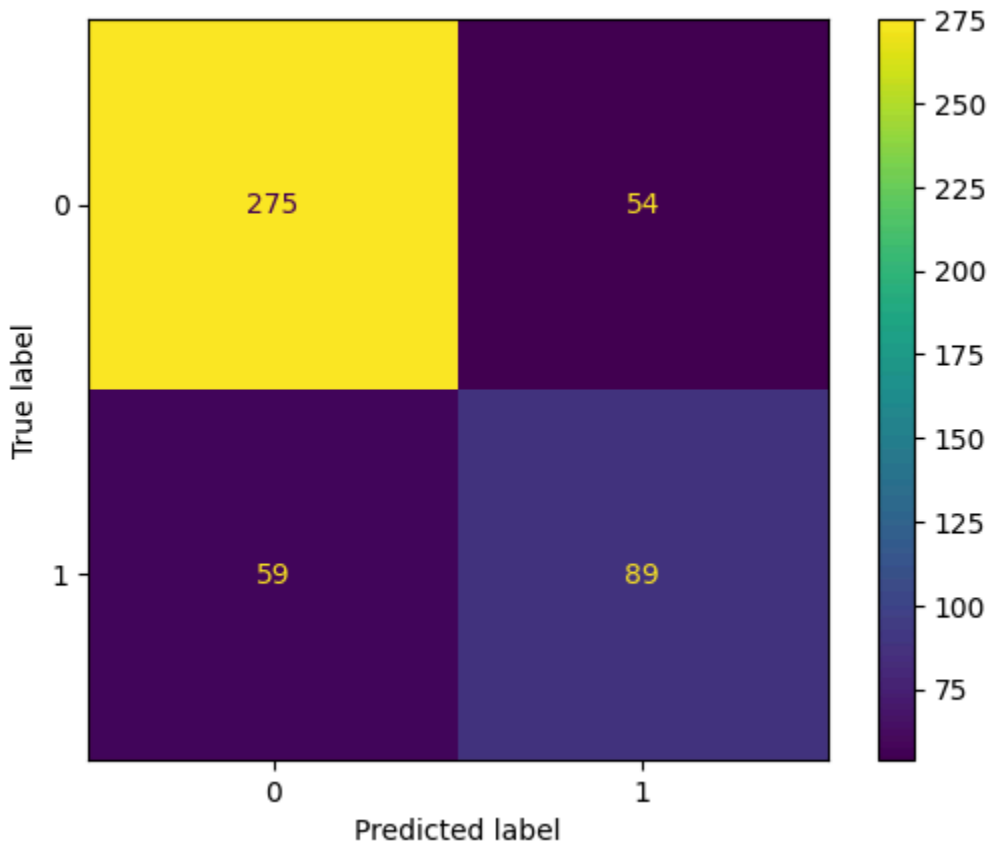
```
import xgboost as xgb

model = xgb.XGBClassifier(class_weight = "balanced")

model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print("XGBoost Classifier Score: ", model.score(X_test, y_test))
print("\n", classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)
ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=model.classes_).plo
```

```
XGBoost Classifier Score:  0.7631027253668763

              precision    recall  f1-score   support

         0.0       0.82      0.84      0.83       329
         1.0       0.62      0.60      0.61       148

    accuracy                           0.76       477
   macro avg       0.72      0.72      0.72       477
weighted avg       0.76      0.76      0.76       477
```

```
/usr/local/lib/python3.12/dist-packages/xgboost/training.py:200: UserWarning:
[14:50:33] WARNING: /__w/xgboost/xgboost/src/learner.cc:782:
Parameters: { "class_weight" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
```

Out[231…

```
<sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7ea82ea37
3e0>
```



XGBoost Classifier with balanced class weight

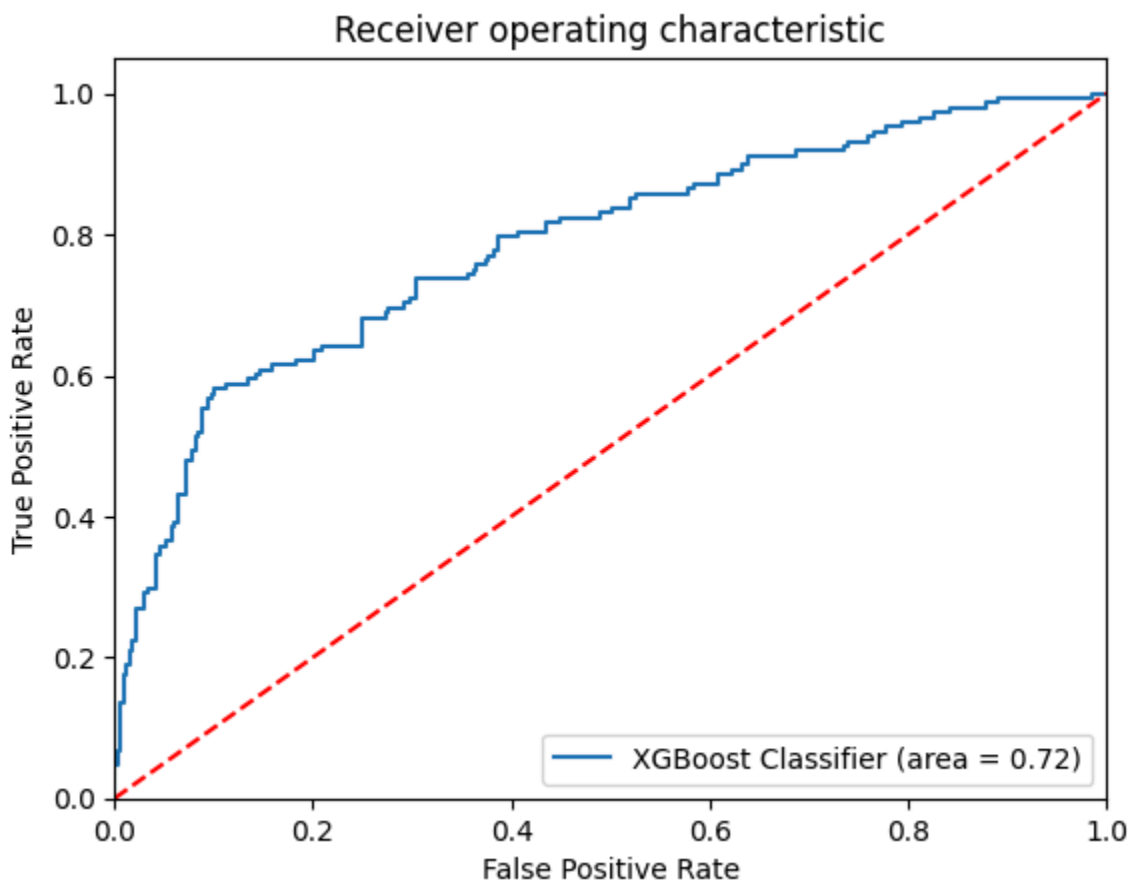
Out of all prediction, the measure for correctly predicted 0 is 62% and for 1 is 81% (Precision) Out of all actual 0, the measure for correctly predicted is 57% and for 1 is 84% (Recall)

As this is imbalanced dataset. We give importance to F1-Score metrics

- F1 Score of 0 is 60%
- F1 Score of 1 is 87%

In [232…

```
logit_roc_auc=roc_auc_score(y_test,y_pred)
fpr,tpr,thresholds=roc_curve(y_test,c.predict_proba(X_test)[:,1])
plt.figure()
plt.plot(fpr,tpr,label='XGBoost Classifier (area = %0.2f)' % logit_roc_auc)
plt.plot([0,1],[0,1],'r--')
plt.xlim([0.0,1.0])
plt.ylim([0.0,1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver operating characteristic')
plt.legend(loc="lower right")
plt.show()
```



Final Result Evaluation:

- We observe that we are not getting very high recall on target 0 which may be due to small unbalanced dataset.
- Higher precision means that an algorithm returns more relevant results than irrelevant ones, and high recall means that an algorithm returns

most of the relevant results (whether or not irrelevant ones are also returned).

- **We observe that Random Forest with SMOTE outperforms rest of the models and has higher recall and precision values.**
    - The Random Forest method out of all predicted 0 the measure of correctly predicted is 73%, and for 1 it is 82%(Precision).
    - The Random Forest method out of all actual 0 the measure of correctly predicted is 56%, and for 1 it is 91%(Recall).
    - The ROC-AUC curve area for Random Forest Classifier is 0.74
- **Gradient Boosting Classifier Result**
    - Out of all prediction, the measure for correctly predicted 0 is 62% and for 1 is 82% (Precision)
    - Out of all actual 0, the measure for correctly predicted is 60% and for 1 is 83% (Recall)
    - The ROC-AUC curve area for Gradient Boosting Decision Tree Classifier is 0.71
- **XGBoost Classifier Result**
    - Out of all prediction, the measure for correctly predicted 0 is 62% and for 1 is 81% (Precision)
    - Out of all actual 0, the measure for correctly predicted is 57% and for 1 is 84% (Recall)
    - The ROC-AUC curve area for XGBoost Classifier is 0.71

Feature Importance of the best model so far.

Random Forest Classifier outperforms the rest of the modal.

Best parameters

Best Params: {'max_depth': 4, 'n_estimators': 50}

In [233…

```
rf = RandomForestClassifier(max_depth = 4, n_estimators= 50, class_weight="bal

rf.fit(X_train, y_train)
print("Score of RandomForestClassifier: ", rf.score(X_test, y_test))
```

```
Score of RandomForestClassifier:  0.80083857442348
```

In [234…

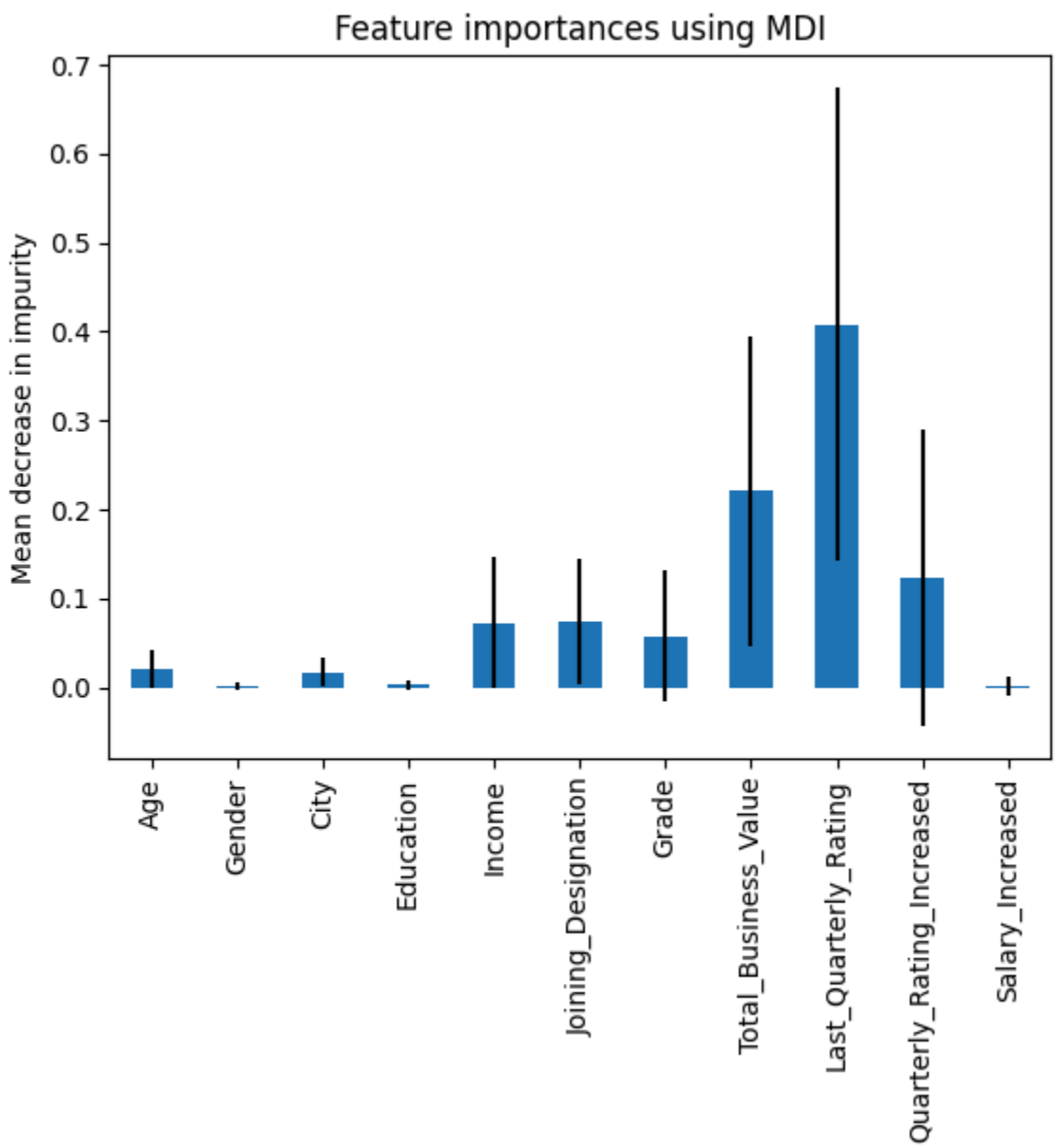
```
importances = rf.feature_importances_
importances
```

Out[234…
```
array([0.02076008, 0.00125778, 0.01738218, 0.00325473, 0.07244162,
       0.07402893, 0.05727424, 0.2208093 , 0.40795498, 0.12255045,
       0.0022857 ])
```

In [235…
```
std = np.std([tree.feature_importances_ for tree in rf.estimators_], axis=0)
```

In [236…
```
feature_importances = pd.Series(importances, X_train.columns)

plt.figure(figsize=(15,7))
fig, ax = plt.subplots()
feature_importances.plot.bar(yerr=std, ax=ax)
ax.set_title("Feature importances using MDI")
ax.set_ylabel("Mean decrease in impurity")

plt.show()
```

```
<Figure size 1500x700 with 0 Axes>
```



Insights:

- Last_Quarterly_Rating, Total_Business_Value & Quarterly_Rating_Increased are the most important features.

Actionable Insights and Recommendation

- Out of 2381 drivers 1616 have left the company.
- We need to incentivise the drivers overtime or other perks to overcome churning
- The employees whose quarterly rating has increased are less likely to leave the organization.
- Company needs to implement the reward system for the customer who provide the feedback and rate drivers
- The employees whose monthly salary has not increased are more likely to leave the organization.
- Company needs to get in touch with those drivers whose monthly salary has not increased and help them out to earn more by provider bonus and perks.
- Out of 2381 employees, 1744 employees had their last quarterly rating as 1.
- Out of 2381 employees, the quarterly rating has not increased for 2076 employees. This is red flag for the company which needs to regulate.
- Company needs to look why customers are not rating drivers.
- Last_Quarterly_Rating, Total_Business_Value & Quarterly_Rating_Increased are the most important features. Company needs to tracks these features as predicators
- We observe that we are not getting very high recall on target 0 which may be due to small unbalanced dataset. More data will overcome this issue.
- The Random Forest Classifier attains the Recall score of 91% for the driver who left the company. Which indicates that model is performing the decent job.